Autonomous Articulated Vehicle Control via Deep Reinforcement Learning and High-Dimensional Perception

by

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Author's Declaration

I hereby declare that I am the sole author of this thesis. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners.

I understand that my thesis may be made electronically available to the public.

Abstract

Autonomous control of articulated tractor-trailer vehicles is difficult because the passive hitch adds an articulation degree of freedom whose reverse dynamics are open-loop unstable, driving the trailer toward the unrecoverable jackknife configuration. This thesis develops and evaluates a learned controller for this setting, targeting the confined low-speed maneuvers typical of distribution-yard and loading-dock operations, and carries it from simulation onto physical hardware.

A Twin Delayed Deep Deterministic Policy Gradient (TD3) agent is trained in a custom Gymnasium simulation of a shunt-truck tractor-trailer. Controlled ablations isolate the learning algorithm, the reward formulation, and the observation modality. A multiplicative reward that jointly penalizes tracking error and hitch articulation is what makes the unstable reverse task learnable, and a compact engineered state proves more reliable than higher-dimensional LiDAR or bird's-eye-view image observations under an identical training recipe. Training a policy directly on the raw image fails, and this failure is shown to be an optimization pathology of off-policy actor-critic learning rather than a shortage of perceptual information. It is resolved by distilling the image into the vehicle's path geometry with a frozen, supervised encoder, over which a policy trains from scratch as readily as on the engineered state. The same distillation extends to obstacle navigation, where the learned controller follows an image-inferred avoidance path and decides when a layout is impassable and the vehicle must stop.

The trained forward and reverse policies are deployed on a physical articulated platform, an AgileX Hunter SE, through a stripped-down Autoware stack, supported by an onboard LiDAR hitch-angle estimator that requires no trailer-mounted sensor and a Smith-predictor compensator for actuator delay.

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List of Abbreviations

BEV	Bird's-Eye View
CAN	Controller Area Network
CNN	Convolutional Neural Network
CTE	Cross-Track Error
DDPG	Deep Deterministic Policy Gradient
DQN	Deep Q-Network
GNSS	Global Navigation Satellite System
GPU	Graphics Processing Unit
IMU	Inertial Measurement Unit
LiDAR	Light Detection and Ranging
LQR	Linear-Quadratic Regulator
MASc	Master of Applied Science
MDP	Markov Decision Process
MPC	Model Predictive Control
MVSL	Mechatronics Vehicle Systems Lab
NDT	Normal Distributions Transform
NMPC	Nonlinear Model Predictive Control
PID	Proportional-Integral-Derivative
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
ROS	Robot Operating System
RTK	Real-Time Kinematic
SAC	Soft Actor-Critic
SDF	Signed-Distance Field
TD3	Twin Delayed Deep Deterministic Policy Gradient
UGV	Unmanned Ground Vehicle

List of Symbols

δ	Tractor steering angle
$\dot{\delta}$	Steering rate (control action)
γ	Hitch articulation angle
ψ	Tractor heading
ψ_t	Trailer heading
$\dot{\psi}$	Tractor yaw rate
$e_y,\ e_\psi$	Tractor cross-track and heading error
$e_{y,t},\ e_{\psi,t}$	Trailer cross-track and heading error
$\kappa_1,\ \kappa_2$	Near and far path-curvature samples
L_h	Signed hitch offset from the tractor rear axle
L_t	Trailer wheelbase (hitch to trailer axle)
$l_f,\ l_r$	Tractor front and rear axle distance from the center of gravity
m	Tractor mass
I_z	Tractor yaw moment of inertia
$C_f,\ C_r$	Front and rear cornering stiffness
v	Longitudinal speed
σ	Stop signal (control action)
d	Geometric obstacle-difficulty rating
γ_d	Reinforcement-learning discount factor
α_k	Guided-reward annealing coefficient
Δt	Simulation timestep

Chapter 1

Introduction

1.1 Motivation

[Section content omitted for review.]

1.2 Problem Definition

[Section content omitted for review.]

1.3 Thesis Contributions

[Section content omitted for review.]

1.4 Institutional Context

[Section content omitted for review.]

1.5 Thesis Organization

[Section content omitted for review.]

Chapter 2

Literature Review

Controlling an autonomous tractor-trailer system can be done through two distinct methods, classical model or optimization based or learning-based planning and control. This chapter reviews both, with the addition of sim-to-real transfer and safety considerations that bound any physical deployment. This review begins with the control problem that makes articulated vehicles distinct, surveys the classical methods that have traditionally addressed it and the learning-based methods that now challenge them, examines how perception, the simulation-to-reality gap, and the governing safety requirements shape a learned policy, and closes by identifying the specific gap that this thesis fills.

2.1 The Articulated-Vehicle Control Problem

Autonomous driving is conventionally decomposed into a pipeline of perception, planning, and control, with each stage feeding the next [1]. Within that pipeline the tractor-trailer is a demanding case for the control stage, because of the passive hitch that couples its two bodies.

Prior analyses have established why. The hitch adds an internal configuration variable, the articulation angle, whose reverse dynamics are open-loop unstable. In forward motion a small hitch disturbance decays as the tractor pulls the trailer back into alignment, but in reverse the sign of the longitudinal velocity flips this behavior. A small articulation error grows rather than decays and, without corrective steering, runs away to the jackknife condition, a folded configuration from which no steering input can recover the trailer [2]. Reverse driving is additionally non-minimum-phase, in that to move the trailer to one side the tractor must first steer to the other [3]. These two properties, open-loop instability and non-minimum-phase response, are what separate articulated control from ordinary rigid-body lane keeping. The kinematic and dynamic model that produces them is derived in

chapter 3, and the confined low-speed setting that fixes the capabilities a controller must deliver is set out in chapter 1.

2.2 Classical Model-Based Control

The longest-established response to articulated instability is model-based control, deriving a dynamic model, then designing a control law that provably stabilizes it. This section reviews the two layers of that approach, the feedback controllers that stabilize and track, and the motion planners that supply the reference for them to track. Together they form the conventional pipeline against which the learned controller of this thesis is benchmarked.

2.2.1 Feedback Control and Stabilization

At the simplest level, a proportional-integral-derivative (PID) controller is trivial to implement and computationally negligible, but it acts only on the present error and cannot anticipate the jackknife condition. Its gains must also be re-tuned for different payloads and path distributions. A linear state-feedback law derived from the linearized tractor-trailer model addresses the multi-variable coupling more directly. Choosing that feedback as a linear-quadratic regulator (LQR) trades tracking accuracy against control effort through a quadratic cost and is optimal for the linearized plant [4]. The author's prior work applied exactly this design to the linearized tractor model [5].

Model predictive control (MPC) generalizes linear feedback by re-optimizing the steering sequence over a receding horizon subject to explicit state and input constraints [6]. This constraint handling is valuable for articulated vehicles because a hard bound can be placed directly on the articulation angle. Linear MPC suffices near an operating point, whereas nonlinear MPC (NMPC) retains the full nonlinear kinematics, including the reverse instability, at a higher computational cost [7]. Control tailored specifically to the articulated case has taken two further forms. Active trailer steering adds an actuated degree of freedom at the trailer axle or hitch to damp articulation directly, and has been shown to improve both the high-speed stability and the low-speed maneuverability of tractor-semitrailers [8,9]. For a conventional trailer with no such actuation, all authority rests on the tractor's steering angle, and recent work has used an intrinsically stable MPC formulation to suppress jackknifing while tracking a reference, exploiting stable inversion of the non-minimum-phase dynamics [10].

2.2.2 Geometric and Nonlinear Control

A parallel line of work exploits the geometric structure of the non-holonomic system directly. Trailer systems are a classic example of a differentially flat system, meaning that the full state and the control inputs can be written algebraically in terms of a flat output and its derivatives, which reduces trajectory generation to the problem of shaping that output [11]. Converting the kinematics to chained form enables time-varying feedback laws with provable path-following and point-stabilization guarantees for wheeled vehicles [12]. For the specific and difficult case of reverse motion, feedback schemes have been derived that stabilize a truck and trailer backing along a path in spite of the open-loop instability [3]. These methods achieve strong guarantees by committing fully to an analytical model of the vehicle, which is precisely the commitment that the learned controller of this thesis seeks to avoid.

2.2.3 Motion Planning for Articulated Vehicles

The feedback controllers above track a reference path, which must first be planned. For articulated vehicles this planning problem is itself difficult, because a feasible path must respect the vehicle's non-holonomic constraints and avoid configurations from which the trailer cannot recover. Sampling-based planners such as the rapidly-exploring random tree and its asymptotically optimal variant RRT* explore the configuration space without an explicit discretization [13, 14], whereas search-based freespace planners such as hybrid A* discretize the search yet retain kinematic feasibility [15]. Tailored to the articulated case, lattice-based planners assemble precomputed motion primitives that connect circular-equilibrium configurations, enabling complex forward and reverse maneuvers [16], and optimization-based formulations compute reference paths that minimize the swept area of the vehicle bodies subject to road and obstacle constraints [17]. Once a path exists, geometric trackers such as pure pursuit follow it by steering toward a look-ahead point on the path [18], and refined trackers such as the Stanley controller add an explicit cross-track error term with demonstrated performance on full-scale autonomous vehicles [19].

2.2.4 Strengths and Limitations

The appeal of this pipeline is that it comes with guarantees. Given an accurate model, the stability of the controller [7, 12] and the feasibility of the planned path [13, 14] can be analyzed and, in principle, guaranteed. Its cost is precisely that dependence. The model must be identified and kept accurate across payloads and operating conditions, the controllers and planners must be tuned, and the system must be assembled from separately designed

planning and control stages. These costs motivate the alternative pursued in this thesis, a controller learned directly from interaction, which maps observations to steering commands without an explicit model inversion or a separately planned reference trajectory, though the path to be followed still enters through the policy's observation, and which is benchmarked in chapter 5 against tuned instances of exactly the classical controllers reviewed here.

2.3 Learning-Based Control

Where a classical controller encodes the vehicle model explicitly and executes a fixed control law, a reinforcement learning (RL) agent infers a control policy directly from interaction with its environment, optimizing the policy end-to-end against a scalar reward [20]. This trades the analytical guarantees of the model-based approach for the ability to learn behaviors that are difficult to hand-design and to fold perception and control into a single learned mapping. This section traces the RL methods relevant to the thesis from their general foundations to their specific application to articulated vehicles.

2.3.1 Deep Reinforcement Learning Foundations

Modern RL for control rests on the deep-network function approximators that made high-dimensional value estimation practical. The Deep Q-Network first showed that a single architecture could learn to act directly from high-dimensional visual input across a range of tasks, using experience replay and a target network to stabilize learning [21]. Value-based methods of this kind are naturally suited to discrete action sets. The continuous steering and speed commands of a vehicle instead call for policy-based methods.

2.3.2 Actor-Critic and Policy-Gradient Methods

The deterministic policy gradient provides a route to continuous control by learning a deterministic actor alongside a critic that estimates action value [22]. Deep Deterministic Policy Gradient (DDPG) is the deep-network realization of this idea and a canonical off-policy actor-critic algorithm for continuous action spaces [23], but it is prone to critic overestimation bias and unstable updates. Two lines of work address this. Proximal Policy Optimization (PPO) is an on-policy method that improves stability through a clipped objective, at the cost of lower sample efficiency, which matters when simulation rollouts are expensive [24]. Soft Actor-Critic (SAC) instead augments the reward with an entropy term to encourage exploration and improve robustness in continuous control [25]. This thesis adopts Twin Delayed DDPG (TD3), a direct successor to DDPG that targets its two weaknesses through twin

critics, delayed actor updates, and target-policy smoothing [26], and whose role is detailed in chapter 4. The author's prior coursework used DDPG for single-body path following [27]. The move to TD3 here is a direct response to the training instability observed in that work.

2.3.3 Reinforcement Learning for Autonomous Driving

RL has been applied across the driving stack, and recent surveys catalog its use for lane keeping, behavioral decision making, and end-to-end control [28]. Two threads are relevant here. The first is end-to-end learning, in which a single network maps sensor input to control, whether trained by imitation of a human driver [29] or by reinforcement. A notable demonstration of this trained a lane-following policy on a full-scale car with on-board RL in a single day [30]. The second is the explicit framing of the driving task as a Markov decision process with a shaped reward, as set out in early RL-for-driving frameworks [31]. Both threads inform the observation and reward design developed in this thesis.

2.3.4 Reinforcement Learning for Articulated Vehicles

Applying RL specifically to articulated vehicles is more recent but growing, and the earliest instance long predates deep RL. The neural truck-backer-upper of Nguyen and Widrow learned to reverse a trailer toward a target, and depended on training that exposed the network to the unstable configurations it had to correct [32]. Modern work has revisited the problem with deep methods. A DDPG backing controller paired with a fuzzy-logic supervisor learned to reverse while avoiding the jackknife state [33], and policy-gradient agents have been trained to track reference paths with a tractor-trailer using rewards that penalize both tracking deviation and articulation growth [34]. A recurring theme in this small body of work, and one this thesis examines directly through its reward and observation ablations, is the importance of the hitch-stability term. The reward must discourage articulation growth for reverse driving to be learned reliably. Reward shaping of this kind rests on a firm theoretical footing, as potential-based shaping can accelerate learning while provably preserving the optimal policy [35].

2.3.5 Sequence Models and Curriculum Learning

Beyond actor-critic control, Decision Transformers recast RL as sequence modeling, predicting an action from a context window of past states, actions, and returns-to-go [36]. Such models can represent longer temporal dependencies than a shallow actor-critic policy, but their performance depends heavily on the diversity of the offline dataset, and a dataset dom-

inated by nominal tracking behavior tends to give poor recovery from the high-error states it underrepresents, an instance of the distribution-shift problem that motivates deliberately collecting such states. This need for coverage of hard configurations reappears as a training-data problem regardless of the policy class, and it motivates curriculum learning: presenting tasks in a deliberately ordered sequence, or automatically emphasizing the situations the agent handles worst [37, 38]. This thesis retains an actor-critic formulation but adopts a failure-replay curriculum (chapter 4) that revisits failed scenarios, an idea related in spirit to prioritized experience replay [39].

2.3.6 Learning under Actuation Delay

A policy trained against an idealized, instantaneous actuator can degrade when deployed on hardware whose steering and drive systems exhibit dead time and first-order lag. A constant actuation delay violates the Markov property of the underlying decision process, turning it into a delayed MDP whose optimal policy depends on the recent action history rather than the instantaneous state alone [40]. The learning literature offers two broad remedies: augmenting the state with a buffer of in-flight actions so that the delayed problem is again Markovian, and correcting the off-policy value estimates for the delay, as in the Delay-Correcting Actor-Critic [41]. This thesis instead takes a complementary, model-based route at deployment, namely a Smith-predictor compensator [42] that presents the delay-free-trained policy with a delay-compensated state estimate (chapter 6), which avoids retraining the policy against the physical delay.

2.4 Perception and Representation Learning

A learned controller is only as good as what it perceives. A central question of this thesis is whether a compact, hand-engineered state vector or a learned representation of a bird's-eye-view (BEV) image yields better and more transferable control. This section reviews the perception building blocks that make the latter possible.

2.4.1 Convolutional Networks for Driving

Convolutional neural networks (CNNs) are the standard front-end for visual perception, learning spatial features directly from images rather than from hand-specified descriptors [43, 44]. Network depth was later shown to be trainable at scale through residual connections, which underpin most modern vision backbones [45]. In driving, CNNs have been trained end-to-end to map camera images directly to steering commands [46]. The bird's-eye view

is a particularly convenient representation for ground-vehicle control because it places the vehicle and its surroundings in a top-down frame. Recent perception systems learn to lift camera images into exactly such a representation [47], and the classical occupancy grid is its long-standing hand-built analogue [48]. In this thesis a CNN processes a rendered BEV image to extract lane geometry and obstacle layout.

2.4.2 Representation Learning and Autoencoders

Training a control policy directly on raw images is sample-inefficient, because the policy must learn to see and to act at the same time. Autoencoders decouple these by compressing images into a compact latent space in an unsupervised pre-training stage [49], an idea applied to RL to shrink the policy's input and accelerate learning [50]. Variational autoencoders impose a probabilistic structure on the latent space that can improve its regularity [51], and spatial autoencoders tailored to control learn latent features corresponding to task-relevant image locations [52]. The U-Net architecture, originally developed for image segmentation, preserves multi-scale spatial detail through skip connections and provides a stronger structural prior than a plain convolutional encoder [53]. This thesis treats these learned encoders as one end of a representation spectrum, with the engineered state vector at the other, and compares them directly in chapter 4.

2.4.3 Range Sensing and Point-Cloud Perception

Bird's-eye imagery is not the only high-dimensional modality available to a ground vehicle. Planar LiDAR provides a direct geometric measurement of the surrounding free space, which the classical occupancy grid represents as a discretized map of drivable space [48]. At the other extreme of complexity, deep networks such as PointNet learn features directly from raw, unordered three-dimensional point clouds [54]. This thesis deliberately adopts a lighter-weight LiDAR representation that sits between these poles, a fixed-length vector of range returns appended to the state vector, rather than a learned point-cloud encoder. This keeps the observation compact and the sensing model simple, and it allows the LiDAR and image modalities to be compared on an equal footing in chapter 4.

2.5 Sim-to-Real Transfer

Because the controllers in this thesis are trained entirely in simulation, transferring a simulation-trained policy to physical hardware is a first-class concern rather than an afterthought. The central obstacle is the reality gap, the systematic mismatch between

simulation and hardware that degrades a naively transferred policy [55]. It is useful to separate this gap into a visual component, a difference in what the sensors report, and a dynamics component, a difference in how the vehicle responds to a command, because the two are attacked by different tools.

The dominant learning-based remedy is domain randomization, which randomizes parameters during training so that the real system appears to the policy as one more variation of a distribution it has already learned to handle. It has been applied to the visual gap by randomizing rendering [56] and to the dynamics gap by randomizing physical parameters such as mass and friction [57]. Randomization can also be closed into a loop. System-identification methods adapt the simulator's parameter distribution from a small amount of real experience, so that simulated and real behavior are matched rather than merely broadened [58], while domain-adaptation methods instead learn a mapping that reconciles simulated and real observations [59]. A complementary strategy is to raise simulator fidelity directly. The open urban driving simulator CARLA is a widely used example [60], and chapter 6 describes a geographically faithful simulator built for this work. Actuation delay is a particularly damaging instance of the dynamics gap for articulated reversing, and is addressed both by the delay-aware learning methods discussed above and by the Smith-predictor compensator of chapter 6. That chapter characterizes the specific gaps encountered on the physical platform, namely actuation delay, hitch-angle sensing noise, and path-distribution mismatch, and addresses them through targeted engineering compensation while noting domain randomization as a route to further robustness.

2.6 Safety and Standards

A controller intended for a physical vehicle must ultimately answer to safety requirements, and the learning-based setting raises safety questions of its own. Safe reinforcement learning studies how to learn while respecting constraints or avoiding catastrophic states [61], and control barrier functions offer a complementary, model-based mechanism to overlay hard safety constraints on a learned policy, a direction identified here as future work [62]. For an articulated vehicle the binding safety concern is the jackknife condition, which the reward must discourage and the operational limits of chapter 6 enforce. A fielded deployment would in addition have to sit within the established automotive-safety framework, most directly the Safety of the Intended Functionality standard for hazards that arise under nominal operation [63], alongside the broader functional-safety, safety-case, and automation-level standards [64–66]. These requirements bound the present simulation study rather than being evaluated within it, and they frame the deployment reported in chapter 6.

2.7 Synthesis and Research Gap

The literature reviewed in this chapter divides into a mature model-based tradition and a fast-moving learning-based one, and the gap this thesis addresses lies between them. Classical control offers guarantees but demands an accurate model, careful tuning, and a separate planning stage. Learning-based control removes those demands but must contend with reward design, the reverse instability, high-dimensional perception, and the reality gap. To our knowledge, prior learned controllers for articulated vehicles have been demonstrated almost exclusively in simulation and on a single, hand-specified observation modality. The backing controllers of Nguyen and Widrow [32] and Bejar and Morán [33], and the tracking agent of Kang et al. [34], each learn from a fixed state representation and report results in simulation, without a physical deployment or a comparison across perception modalities. This thesis addresses that combination directly. It develops a learned articulated-vehicle controller with high-dimensional perception, compares observation modalities and reward formulations through controlled ablations, benchmarks the result against tuned classical baselines, and deploys the trained policies on physical hardware with explicit attention to hitch stability and sim-to-real transfer. The remainder of the thesis carries out this program, beginning with the vehicle model and simulation environment in chapter 3.

Chapter 3

Modeling, Simulation, and Problem Formulation

3.1 Tractor Dynamics

The tractor is represented by a planar single-track bicycle model [67], in which the front axle is lumped into one steerable wheel and the rear axle into one fixed wheel. Its global position (x, y) follows from the body-frame longitudinal and lateral velocities (v_x, v_y) and the heading ψ:

$$\dot{x} = v_x \cos\psi - v_y \sin\psi \tag{3.1}$$

$$\dot{y} = v_x \sin\psi + v_y \cos\psi \tag{3.2}$$

Lateral force is generated at each tire as a function of its slip angle. Under the small-angle assumptions appropriate to the operating envelope, the front and rear slip angles are

$$\alpha_f = \delta - \frac{v_y + l_f\dot{\psi}}{v_x}, \qquad \alpha_r = -\frac{v_y - l_r\dot{\psi}}{v_x}, \tag{3.3}$$

where δ is the steering angle. Rather than evaluating the full Pacejka magic-formula tire curve, the simulator operates in its linear region. For the modest slip angles encountered here the curve is well approximated by a constant cornering stiffness, giving the lateral tire forces

$$F_{yf} = C_f\,\alpha_f, \qquad\qquad F_{yr} = C_r\,\alpha_r. \tag{3.4}$$

This linearized-Pacejka (constant cornering-stiffness) model is the one implemented in the simulator. It preserves the lateral behavior relevant to path tracking while avoiding the hard-to-identify parameters of the full nonlinear tire curve. Collecting the longitudinal, lateral, and yaw-moment balances, with a quadratic aerodynamic drag $F_{\text{drag}} = \frac{1}{2}C_d\rho A v_x^2$ and a net drive force F_x, yields the tractor dynamics. Although the drag force is negligible at low speeds, it was added as a damping force so that the simulation eventually came to rest under no input. Active braking is achieved through applying a negative drive force.

$$m\left(\dot{v}_x - v_y\dot{\psi}\right) = F_x - F_{\text{drag}} - F_{yf}\sin\delta, \tag{3.5}$$

$$m\left(\dot{v}_y + v_x\dot{\psi}\right) = F_{yf}\cos\delta + F_{yr}, \tag{3.6}$$

$$I_z\,\ddot{\psi} = l_f F_{yf}\cos\delta - l_r F_{yr}. \tag{3.7}$$

For control design and for the time-stepping used in simulation, these equations are linearized about straight-line motion ($\delta \approx 0$) at a constant reference speed $v_x = v$. With state $\boldsymbol{\xi} = [\,v_x,\ v_y,\ \dot{\psi}\,]^\top$ and input $\mathbf{u} = [\,F_x,\ \delta\,]^\top$, the linearized dynamics take the time-invariant form $\dot{\boldsymbol{\xi}} = A\boldsymbol{\xi} + B\mathbf{u}$ with

$$A = \begin{bmatrix} -\dfrac{C_d\rho A v}{m} & 0 & 0 \\ 0 & -\dfrac{C_f + C_r}{mv} & \dfrac{l_r C_r - l_f C_f}{mv} \\ 0 & \dfrac{l_r C_r - l_f C_f}{I_z v} & -\dfrac{l_f^2 C_f + l_r^2 C_r}{I_z v} \end{bmatrix}, \qquad B = \begin{bmatrix} \dfrac{1}{m} & 0 \\ 0 & \dfrac{C_f}{m} \\ 0 & \dfrac{l_f C_f}{I_z} \end{bmatrix}. \tag{3.8}$$

The continuous system is converted to the discrete update $\boldsymbol{\xi}_{k+1} = A_d\boldsymbol{\xi}_k + B_d\mathbf{u}_k$ by a zero-order-hold discretization at the simulation step $\Delta t = 0.1\,\text{s}$, and it is this discrete map that is advanced at each timestep.

This level of fidelity is sufficient for the control tasks studied in this thesis, where the quantities of interest are path-relative position error, heading error, steering angle, and hitch articulation. The aim is a physically meaningful environment for training and evaluating classical and trained policies, not a high-fidelity vehicle simulator. An overly simplified kinematic model of the tractor would make the simulation task unrealistically easy and could reduce transferability to physical systems by neglecting actuator response, steering-

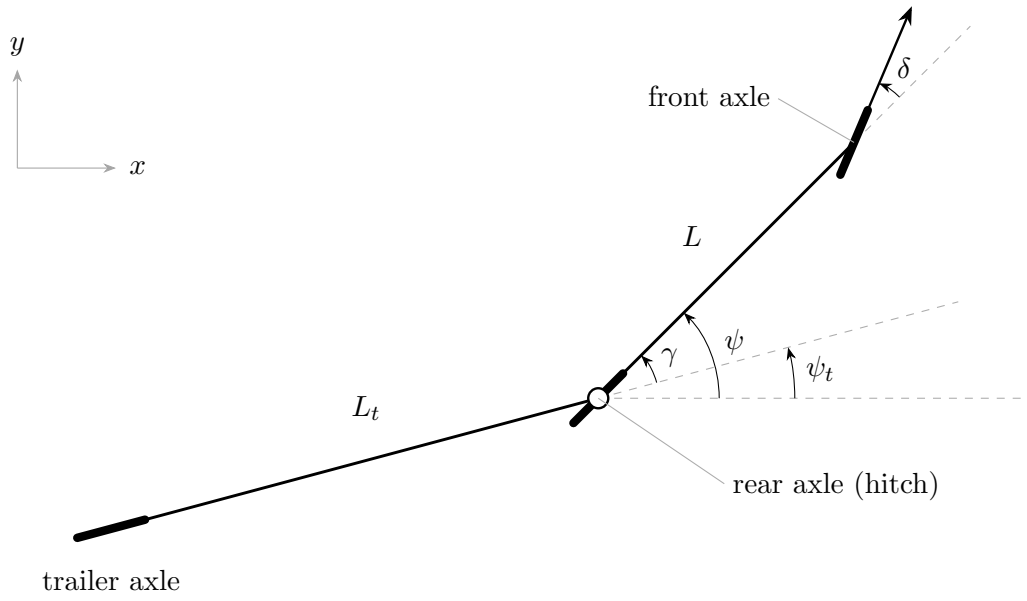


Figure 3.1: Tractor-trailer geometry. The tractor rear axle carries the hitch ($L_h \approx 0$), from which the trailer of wheelbase L_t trails at articulation angle $\gamma = \psi - \psi_t$. The front wheel is steered by δ relative to the tractor heading ψ, and L is the tractor wheelbase.

rate limits, and tire-force effects on the vehicle's motion. In contrast, a kinematic model was chosen for the trailer because, at the low speeds considered in this thesis, the trailer's dominant behavior is governed by its rolling constraint rather than by lateral tire-slip or inertial dynamics. The full-length trailer therefore primarily contributes geometric articulation through the hitch. Its axle constrains the trailer to roll along its longitudinal direction, and this nonholonomic constraint determines the evolution of the hitch angle during forward and reverse maneuvers. This reduced-order representation preserves the main source of reverse instability while avoiding the additional tire, load-transfer, and yaw-inertia parameters required by a full dynamic trailer model.

3.2 Trailer Articulation and Reverse Instability

Figure 3.1 illustrates the tractor-trailer geometry, the tractor and trailer headings ψ and ψ_t, and the articulation angle $\gamma = \psi - \psi_t$ used throughout this section.

The articulated system adds a trailer body coupled to the tractor through a revolute hitch. The relevant geometry is:

- L_h: signed distance from the tractor rear axle to the hitch point (in both the training model and the hardware platform the hitch sits at the rear axle, so $L_h \approx 0$);
- L_t: distance from the hitch to the trailer axle, i.e. the trailer wheelbase;

- $\gamma = \psi - \psi_t$: the articulation angle between the tractor heading ψ and the trailer heading ψ_t (the same convention used by the deployment in chapter 6).

The hitch point is carried rigidly by the tractor,

$$x_h = x - L_h \cos\psi, \qquad y_h = y - L_h \sin\psi, \tag{3.9}$$

and translates with the tractor's longitudinal speed v. Imposing a pure-rolling (no lateral slip) constraint on the trailer axle fixes the trailer yaw rate from the lateral component of the hitch velocity:

$$\dot{\psi}_t = \frac{v}{L_t} \sin(\psi - \psi_t) = \frac{v}{L_t} \sin\gamma. \tag{3.10}$$

The articulation angle therefore evolves as

$$\dot{\gamma} = \dot{\psi} - \frac{v}{L_t} \sin\gamma, \tag{3.11}$$

where the tractor yaw rate $\dot{\psi}$ is supplied by the tractor model of eq. (3.8) (equivalently $\dot{\psi} = (v/L)\tan\delta$ for a purely kinematic tractor of wheelbase $L = l_f + l_r$). The trailer axle pose then follows from the hitch point and the trailer heading,

$$x_t = x_h - L_t \cos\psi_t, \qquad y_t = y_h - L_t \sin\psi_t. \tag{3.12}$$

The sign of v in eq. (3.11) is what makes reverse motion qualitatively harder. Linearizing about the aligned configuration ($\gamma \approx 0$, so $\sin\gamma \approx \gamma$) and holding the steering-driven tractor yaw fixed, the unforced articulation mode obeys

$$\dot{\gamma} \approx -\frac{v}{L_t}\,\gamma, \qquad \lambda = -\frac{v}{L_t}. \tag{3.13}$$

In forward motion ($v > 0$) the eigenvalue λ is negative. Articulation errors decay and the trailer self-aligns behind the tractor. In reverse ($v < 0$) the eigenvalue becomes positive and the same errors grow exponentially toward the jackknife condition. This is an open-loop instability rather than a mere tuning difficulty, and it is the defining feature of reverse articulated maneuvering, the reason the reverse task demands active hitch stabilization, trailer-referenced error metrics, and the dedicated reward and curriculum treatments developed in chapter 4.

The jackknife toward which this instability drives is a folded configuration from which no steering input can recover the trailer's alignment, so successful reverse maneuvering demands anticipatory control that corrects a developing articulation error before it grows large enough

to become unrecoverable. In practice the controller must regulate three coupled objectives at once (trailer path tracking, tractor placement, and articulation stability), and a controller that attends only to the tractor's cross-track error can look reasonable in the short term while driving the hitch toward instability.

3.3 Vehicle Parameters

The vehicle model is parameterized to represent a shunt truck (terminal tractor, or yard hostler), the class of vehicle that performs the confined low-speed trailer maneuvering in distribution centers that this thesis targets. The specific reference vehicle is the Kalmar Ottawa T2 4x2 terminal tractor. Its layout maps directly onto the single-track bicycle model of eq. (3.8), and its manufacturer specification sheet provides genuine primary-source data [68]. The tractor has a wheelbase of 2.95 m, a maximum steering angle of 50° (0.87 rad), and is governed to a top speed near 20 m/s, more than enough to operate in the low-speed regime studied here. Table 3.1 lists the parameter set used in simulation, corresponding to the bobtail (no-trailer) diesel configuration.

Table 3.1: Vehicle parameters used in simulation, for the bobtail Kalmar Ottawa T2 diesel terminal tractor.

Parameter	Value	Parameter	Value
m	6577 kg	l_f	1.33 m
I_z	16 000 kg m^2	l_r	1.62 m
C_f	190 000 N/rad	C_d	0.8
C_r	200 000 N/rad	A	7.0 m^2
Δt	0.1 s	ρ	1.225 kg/m^3

The mass, wheelbase, and overall dimensions are taken directly from the manufacturer specification sheet. The yaw moment of inertia I_z is estimated from the mass and body envelope using a standard moment-of-inertia estimator for road vehicles [69]. The cornering stiffnesses C_f and C_r are derived from published heavy-truck tire data [70], scaled by the number of tires per axle (a single pair at the steered front axle against dual tires at the drive axle) and by the axle loads. The front and rear stiffnesses therefore differ, unlike the equal-stiffness simplification commonly used for passenger-car examples. The longitudinal center-of-gravity split is estimated from an approximate static front/rear axle load distribution, and the aerodynamic terms C_d and A are engineering estimates for a terminal-tractor cab.

Neither the aerodynamic terms nor the center-of-gravity split is backed by primary data, but their influence is minor in the operating regime of interest. Aerodynamic forces are negligible at governed yard speeds, and the center-of-gravity split mainly shapes high-speed transient response rather than the low-speed trajectories studied here. The timestep $\Delta t = 0.1\,\mathrm{s}$ corresponds to the 10 Hz control loop used in the hardware deployment (chapter 6), so the simulation and the deployed controller operate at a matched rate.

The values in table 3.1 describe the bobtail tractor, which defines the dynamic tractor subsystem used in simulation. The trailer is represented kinematically through the hitch model of eq. (3.11), for the reasons given in section 3.1. Its inertia is deliberately omitted rather than approximated. Because the tasks prescribe longitudinal speed, trailer mass never enters the control loop through the longitudinal balance, and both the rolling constraint of eq. (3.10) and the reverse-instability eigenvalue of eq. (3.13) are independent of mass and moment of inertia. Trailer inertia is therefore not a control-critical state in the nominal environment. The one regime in which a heavily loaded trailer does reshape the dynamics, through kingpin load transfer onto the tractor, is examined separately as a robustness variant, in which that load is folded into the tractor's mass, yaw inertia, cornering stiffness, and center-of-gravity split rather than introduced as a separate trailer body.

3.4 Simulation Environment

The training and evaluation environment is a custom 2D simulation built on the Farama Foundation Gymnasium library [71]. Gymnasium provides a standardized API for environments to integrate directly with reinforcement learning algorithm libraries, including Stable Baselines3 [72].

The simulation is implemented in Python, with Pygame [73] providing collision masks and rendering. It was built around a rendering pipeline that draws the tractor-trailer and lane geometry as a top-down frame. This representation can be viewed live for monitoring, or passed directly to the image encoder as a bird's-eye-view (BEV) observation, making high-dimensional perception available to the agent without a separate image-synthesis step. At each timestep, the discrete-time tractor-trailer state-space equations are re-evaluated from the previous state and control inputs, updating the vehicle position, heading, hitch angle, and associated error signals, and Pygame redraws the scene as a 2D top-down view from which the BEV image observation is cropped. For large-scale training this rendered observation is later produced without the Pygame renderer by the batched GPU simulator (section 4.9), but the rendering pipeline remains the conceptual origin of the BEV observation and the reference for what it encodes.

The simulation environment includes:

1. **Vehicle Model**: A combined dynamic tractor (bicycle model with a linearized Pacejka tire model [74]) and kinematic trailer with hitch constraints, using the shunt-truck parameters of section 3.3.

2. **Lane Generation**: Randomly generated lane paths using cubic spline interpolation, producing varying curvature paths for training and evaluation.

3. **Obstacle Placement and Local Re-planning**: Randomly placed static obstacles within the lane and a local path planner that perturbs the centerline around those obstacles for avoidance experiments.

4. **Collision Detection**: Pygame sprite mask-based collision detection, providing pixel-precise contact sensing between the vehicle body and obstacles or lane boundaries.

5. **Observations**: the rendered top-down BEV image described above, alongside engineered state and range-sensor (LiDAR) observations. The full set of observation modalities available to the policy is defined in chapter 4.

3.5 Environment Tasks

The environment implementation is structured around reusable task and observation components. A base lane-following task is combined with different observation heads, while obstacle-aware variants reuse the same task logic through an environment that adds obstacle generation, local path planning, and obstacle-aware reward evaluation. This avoids maintaining separate forward, reverse, LiDAR, and BEV implementations with duplicated task logic.

3.5.1 Lane Following

The lane-following environment extends the base tractor-trailer environment to present a lane-tracking task. A reference lane is generated at the start of each episode using cubic spline interpolation through randomly placed waypoints. The reward function encourages the agent to keep both the tractor and trailer within the lane boundaries while maintaining progress along the path.

Reverse lane-following environments use the same path generation and rendering pipeline, but reverse the longitudinal command and compute reward and tracking terms with the

trailer as the leading unit, so that the same task can be studied in backward maneuvers without changing the task definition.

3.5.2 Obstacle Avoidance

The obstacle avoidance environment adds static obstacles within the lane. The agent must navigate the tractor-trailer through the lane while avoiding obstacles, which requires both path-following behavior and collision avoidance. Instead of following the global lane centerline directly, the environment constructs a local path around the placed obstacles and evaluates tracking error relative to this locally planned path. Obstacles are placed as a single cluster of one or two circular obstacles per episode, drawn from a small set of layout categories (clear, shift, squeeze, and blocked), so that the difficulty distribution is controlled and can be stratified during evaluation. The locally planned avoidance path is used only to shape the reward and is never exposed to the agent, so the detour must be inferred from the LiDAR or BEV observation. In addition to terminal collision penalties, the base environment applies a per-step proximity penalty near blocked occupancy-grid cells, so both lane boundaries and inserted obstacles discourage the agent before contact occurs. Forward and reverse obstacle environments are both available. The scalar difficulty rating that indexes these layouts, and the reward that keys on it, are developed in section 4.3.1.

3.5.3 Actions and Episode Termination

The policy is exposed to a deliberately narrow interface. In the reinforcement-learning sense its action is two-dimensional, a continuous steering rate $\dot{\delta}$ and a stop signal σ:

$$\mathbf{a} = [\dot{\delta},\, \sigma] \in [-25^\circ/\mathrm{s},\, 25^\circ/\mathrm{s}] \times [-1,\, 1]. \tag{3.14}$$

Longitudinal speed is not itself a control degree of freedom. The environment is driven by a target speed, but the policy does not set it directly. Instead, a thin interface layer maps the stop signal onto that speed, commanding the fixed operating value of the current task ($5\,\mathrm{m/s}$ forward, $-5\,\mathrm{m/s}$ in reverse) while the vehicle drives and zero speed the instant a stop is issued. Because speed is prescribed in this way, every method solves the same lateral problem and the forward and reverse tasks share a common control interface. Commanding the steering rate rather than the steering angle reflects the fact that the steering angle cannot change instantaneously on a physical vehicle, so regulating its rate of change is more physically realistic. The stop signal is a go/stop decision. When σ exceeds a fixed threshold the vehicle halts and the episode terminates, which lets the policy decline to attempt a

layout it judges impassable. This stop-capable action is used across all learning-based tasks, so that the interface does not change between the no-obstacle and obstacle settings. Discrete algorithms realize the same choice through a joint steering-and-stop action set. Stopping is not universally desirable. In the obstacle-free tasks a stop is treated as a failure and penalized, and it earns reward only on difficult obstacle layouts. The difficulty-keyed stop-gate reward that governs this trade-off is developed in section 4.3.1, and the experimental protocols in chapter 5.

Episodes terminate on collision, jackknife ($|\gamma| > 90°$), or successful path completion.

Chapter 4

Learning Framework: Observation and Representation Design, Reward Formulation, and Policy Training

4.1 Learning Framework Overview

This chapter discusses the development of the learned controller and the supporting framework required to train it. The core learning system is defined first: a Twin Delayed Deep Deterministic Policy Gradient (TD3) agent that operates on a compact engineered state and is driven by a reward that balances path tracking against hitch stability. The higher-dimensional observation modalities, planar LiDAR and the bird's-eye-view image, and the obstacle-aware navigation task are then introduced as extensions of that core system. Because inserting raw perception into an otherwise successful policy turns out to destabilize learning, a dedicated section then develops the training methodology that makes high-dimensional-observation policies trainable at all, working through three progressively cleaner ways to decouple perception from control and arriving at a geometry-distillation encoder that recovers from-scratch learning entirely. The evaluation methodology and the GPU-parallel simulation infrastructure that make the empirical study feasible close the chapter. This progression follows the order in which the design was developed. The algorithm and reward are settled on the engineered state where learning is fast and unambiguous, then the added complexity of raw perception and the training machinery it demands is introduced.

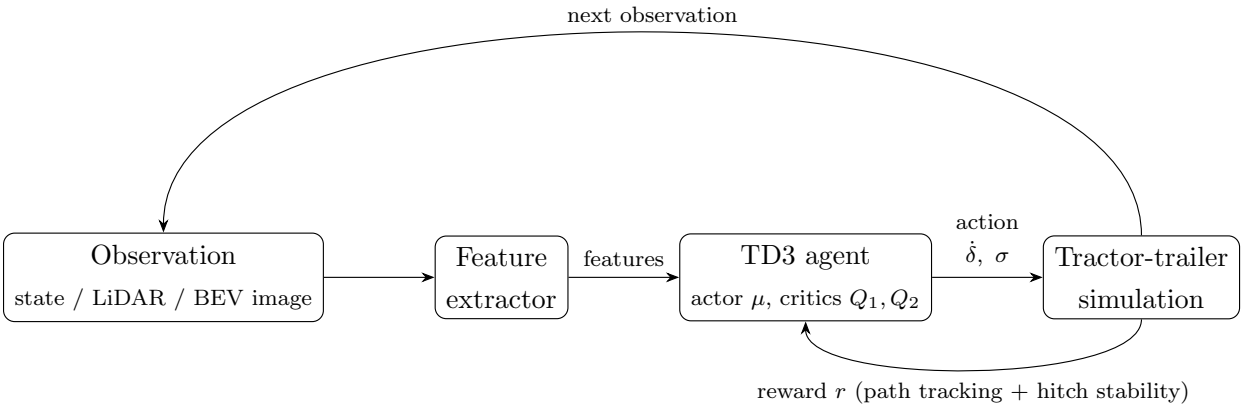


Figure 4.1: Architecture of the learning system: an observation (engineered state, LiDAR, or BEV image) is encoded into a feature vector and passed to the TD3 actor and critics, which output a steering rate and a stop signal. The reward combines path-tracking and hitch-stability terms.

The agent perceives its surroundings through one of three observation families, forming a deliberate progression from a fully engineered representation to raw imagery:

- **State**: an 8-dimensional vector $[\delta, \gamma, e_y, e_\psi, e_{y,t}, e_{\psi,t}, \kappa_1, \kappa_2]$ containing the steering angle, the hitch articulation angle, the tractor and trailer cross-track and heading errors, and two local path-curvature samples.
- **Minimal State + LiDAR**: a 2-dimensional vector $[\delta, \gamma]$ containing the steering angle and the hitch angle augmented with N normalized range readings, where the beam count $N \in \{8, 24, 64\}$ is varied in the LiDAR ablation of chapter 5.
- **Minimal State + BEV image**: The same minimal state vector and a 32×32 grayscale bird's-eye-view image.

The core learning system of the next three sections operates on the engineered state. The LiDAR and image modalities are taken up in section 4.5. This progression lets the marginal value of richer perception be measured against its added representation-learning and inference cost rather than assumed.

The LiDAR- and BEV-based observation spaces are augmented with the minimal intrinsic state vector because steering and hitch angles are available through onboard sensing and state estimation and therefore do not need to be inferred by the exteroceptive perception system.

4.2 Reinforcement Learning with TD3

Twin Delayed Deep Deterministic Policy Gradient (TD3) is an off-policy actor-critic algorithm designed for continuous action spaces. TD3 extends deterministic policy-gradient methods with three key improvements: twin critic networks (to reduce overestimation bias), delayed policy updates (to improve stability), and target policy smoothing (to reduce variance in critic updates). These properties make TD3 a natural fit for articulated-vehicle control: the action space is continuous (a continuous steering rate and a stop signal that is triggered above a threshold), a deterministic policy avoids the action-distribution spread that can aggravate the non-minimum-phase reverse dynamics, and the off-policy replay buffer amortizes the comparatively high cost of image-based rollouts by reusing every transition many times.

The TD3 actor network $\mu(s \mid \theta^{\mu})$ maps states to deterministic actions, while the critic networks $Q_1, Q_2(s, a \mid \theta^{Q_1}, \theta^{Q_2})$ evaluate action quality. The target value is computed using the minimum of the two critic estimates:

$$y = r_t + \gamma_d \min_{i=1,2} Q_i(s_{t+1},\, \mu(s_{t+1}) + \epsilon) \tag{4.1}$$

where $\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c)$ is clipped Gaussian noise added to the target action for smoothing. Parameters are updated by minimizing the temporal difference error in the critics and maximizing the expected return through the actor using the deterministic policy gradient theorem [22].

The TD3 implementation used for this work is from the Stable Baselines3 library [72], which provides a reliable implementation integrated with the Gymnasium environment API [71].

4.3 Reward Formulation

The thesis evaluates several reward variants rather than a single scalar shaping rule. Let e_y and e_ψ denote the tractor cross-track and heading errors, $e_{y,t}$ and $e_{\psi,t}$ the trailer cross-track and heading errors, γ the hitch articulation angle, $\dot{x}$ the longitudinal speed, and δ the steering angle. Longitudinal speed is not a control degree of freedom: it is held at the fixed operating value from chapter 3 ($\dot{x} = \pm 5\,\text{m/s}$ forward or reverse), so $\dot{x}$ appears in the reward only as a near-constant progress term that a stop forfeits, and the second action dimension is a stop signal rather than a commanded speed (chapter 3). The pure-pursuit guide steering is denoted δ_{pp}, with steering deviation $\Delta_\delta = (\delta - \delta_{\text{pp}})/(2\delta_{\max})$. The lane-following base environment also defines a proximity penalty P_{prox} that is subtracted from non-terminal

step rewards. This penalty is zero when the vehicle is farther than $d_{\text{prox}} = 2\,\text{m}$ from blocked occupancy-grid cells and increases quadratically to $P_{\max} = 5$ as clearance approaches zero:

$$P_{\text{prox}} = \begin{cases} 0, & c \geq d_{\text{prox}}, \\ P_{\max}\left(1 - \frac{c}{d_{\text{prox}}}\right)^2, & c < d_{\text{prox}}, \end{cases} \tag{4.2}$$

where c is the minimum clearance between the rendered vehicle footprint and the lane-boundary/obstacle occupancy grid. In the no-obstacle lane-following environment this discourages driving near lane boundaries; obstacle environments inherit the same term for both lane boundaries and inserted obstacle cells.

Forward lane-following rewards

For forward driving, the terminal reward is

$$R_{\text{term}}^{\text{fwd}} = \begin{cases} 100, & \text{success}, \\ -100, & \text{failure}, \end{cases} \tag{4.3}$$

and the common shaping terms are

$$r_p^{\text{fwd}} = 0.5\,\dot{x}, \tag{4.4}$$

$$P_{\text{tractor}} = e_y^2 + e_\psi^2, \tag{4.5}$$

$$P_{\text{trailer}} = (0.5\,e_{y,t})^2 + (0.5\,e_{\psi,t})^2. \tag{4.6}$$

The four forward reward variants are then

$$R_{\text{dense}}^{\text{fwd}} = r_p^{\text{fwd}} - P_{\text{tractor}} - P_{\text{trailer}} - P_{\text{prox}}, \tag{4.7}$$

$$R_{\text{tractor}}^{\text{fwd}} = r_p^{\text{fwd}} - P_{\text{tractor}} - P_{\text{prox}}, \tag{4.8}$$

$$\begin{aligned} R_{\text{mult}}^{\text{fwd}} &= 4\exp(-|e_y| - 0.5|e_\psi| - 0.75|e_{y,t}| - 0.5|e_{\psi,t}|)\exp(-1.5|\gamma|) \\ &\quad + 0.5\,\text{clip}(\dot{x}, 0, 1) - P_{\text{prox}}, \end{aligned} \tag{4.9}$$

$$R_{\text{guide}}^{\text{fwd}}(k) = \alpha_k\left(-5\Delta_\delta^2 - P_{\text{prox}}\right) + (1 - \alpha_k)\,R_{\text{mult}}^{\text{fwd}}. \tag{4.10}$$

The guided reward interpolates between a pure-pursuit imitation term $-5\Delta_\delta^2$, which rewards matching the pure-pursuit steering, and the multiplicative reward $R_{\text{mult}}^{\text{fwd}}$ defined above. The schedule $\alpha_k \in [0, 1]$ governs the transition from imitation toward the task reward and is described in section 4.6, where the same mechanism is reused to fine-tune an imitated policy. At $\alpha_k = 1$ the guided reward is a pure-pursuit clone.

Reverse lane-following rewards

For reverse driving, the terminal reward is

$$R_{\text{term}}^{\text{rev}} = \begin{cases} 200, & \text{success}, \\ -500, & \text{failure}, \end{cases} \tag{4.11}$$

The terminal rewards for the reverse scenario are significantly larger magnitude than the forward variant to encourage the policy to continue driving and learn to stabilize the hitch angle while tracking the path. Using the same terminal rewards as the forward scenario resulted in reverse policies that intentionally ended each episode early with failure to avoid the penalty from poor path tracking that accumulated over each step of the episode.

The shared shaping terms are

$$r_p^{\text{rev}} = 3\,\text{clip}(-\dot{x}, 0, 1), \tag{4.12}$$

$$P_{\text{path}} = e_y^2 + 0.5\,e_\psi^2 + 0.5\,e_{y,t}^2 + 0.25\,e_{\psi,t}^2, \tag{4.13}$$

$$P_{\text{jack}} = 10\max(0, |\gamma| - 0.4)^2. \tag{4.14}$$

The reverse variants are

$$R_{\text{dense}}^{\text{rev}} = r_p^{\text{rev}} - P_{\text{path}} - P_{\text{jack}} - P_{\text{prox}}, \tag{4.15}$$

$$R_{\text{no-hitch}}^{\text{rev}} = r_p^{\text{rev}} - P_{\text{path}} - P_{\text{prox}}, \tag{4.16}$$

$$\begin{aligned} R_{\text{mult}}^{\text{rev}} = {} & 5\exp(-|e_y| - 0.5|e_\psi| - 0.75|e_{y,t}| - 0.5|e_{\psi,t}|)\exp(-2|\gamma|) \\ & + 0.5\,\text{clip}(-\dot{x}, 0, 1) - P_{\text{prox}}, \end{aligned} \tag{4.17}$$

$$R_{\text{guide}}^{\text{rev}}(k) = \alpha_k\left(-5\Delta_\delta^2 - P_{\text{prox}}\right) + (1 - \alpha_k)\,R_{\text{mult}}^{\text{rev}}. \tag{4.18}$$

Obstacle difficulty

The obstacle-aware terms below are keyed to a scalar difficulty $d \in [0, 1]$, a static geometric clearance ratio computed as the tightest cross-section in the observable path window:

$$d = \text{clip}\left(1 - \frac{g_{\max} - w_v}{w_L - w_v},\ 0,\ 1\right), \tag{4.19}$$

where $g_{\max}$ is the widest contiguous free lateral gap, w_v the vehicle width, and w_L the lane width, so $d = 0$ is a fully open lane and $d = 1$ a gap at or below the vehicle width. Its rationale and per-direction calibration are developed in section 4.3.1 and chapter 5.

Obstacle-aware extensions

Obstacle environments inherit the underlying forward or reverse reward, including the proximity penalty above. A failed obstacle episode receives the standard terminal penalty (chapter 3). A deliberate stop instead receives a difficulty-scaled terminal reward,

$$s(d) = \frac{e^{k_d d} - 1}{e^{k_d} - 1}, \qquad k_d = 3, \tag{4.20}$$

$$R^{\text{obs}}_{\text{stop}} = s(d)\, B_{\text{hard}} - \big(1 - s(d)\big) B_{\text{easy}} + p\, B_{\text{prog}}, \tag{4.21}$$

where $p \in [0, 1]$ is the progress made before stopping and $B_{\text{hard}} = B_{\text{easy}} = 60$, $B_{\text{prog}} = 20$. The soft difficulty gate $s(d)$ makes stopping pay on an impassable layout, where it must beat a crash, and lose to driving through on an easy one.

Branching hidden-path reward

Rather than tracking the single hidden avoidance path of chapter 3, the obstacle reward tracks two hidden corridors, one passing each side of the obstacle, and keeps the more favorable of the two evaluations:

$$R^{\text{branch}} = \max_{c \in \{\text{L}, \text{R}\}} R\Big(e_y^{(c)},\, e_\psi^{(c)},\, e_{y,t}^{(c)},\, e_{\psi,t}^{(c)}\Big), \tag{4.22}$$

where $e_\bullet^{(c)}$ are the tracking errors relative to corridor c and $R(\cdot)$ is the underlying forward or reverse reward above. Its rationale is developed in section 4.3.1.

The variants above differ chiefly in how the tracking and articulation terms combine, additively or as a product that forces the errors down jointly rather than letting the agent collect partial credit from a single term, and are compared empirically in chapter 5. Table 4.1 summarizes them.

Table 4.1: Reward variants compared in chapter 5. Every variant also includes the progress term r_p and subtracts the proximity penalty P_{prox}, with terminal rewards as defined above. The variants differ in how the tracking and articulation terms combine.

Variant	Tracking penalty	Articulation term	Role
Forward			
Dense	Additive, tractor + trailer	None	Baseline
Tractor-focus	Additive, tractor only	None	Drops trailer term
Multiplicative	Multiplicative product	$\exp(-1.5\lvert\gamma\rvert)$ factor	Chosen recipe
Guided (PP)	Multiplicative (via $R^{\text{fwd}}_{\text{mult}}$)	Via $R^{\text{fwd}}_{\text{mult}}$	Anneals from pure-pursuit imitation
Reverse			
Dense	Additive (P_{path})	P_{jack}	Baseline
No-hitch	Additive (P_{path})	None	Drops jackknife term
Multiplicative	Multiplicative product	$\exp(-2\lvert\gamma\rvert)$ factor	Chosen recipe
Guided (PP)	Multiplicative (via $R^{\text{rev}}_{\text{mult}}$)	Via $R^{\text{rev}}_{\text{mult}}$	Anneals from pure-pursuit imitation

4.3.1 Obstacle-Navigation Reward Design

The obstacle-navigation task reuses the lane-following reward but adds three design elements, defined in the equations above and motivated here: a difficulty rating, a stop-gate, and a branching hidden-path reward. The obstacle environment itself was described in chapter 3.

Difficulty as a geometric proxy

The difficulty d of eq. (4.19) is a deliberately cheap, static geometric proxy for drivability, used because evaluating true kinematic feasibility for a non-holonomic tractor-trailer inside the training loop would be far too slow. Measuring only whether the body width fits the gap has two consequences that matter later: every gap at or below the vehicle width collapses to $d = 1$, and, because the proxy ignores the maneuvering and trailer-swing room a real trajectory needs, it labels forward and reverse layouts identically even though reverse needs far more room. Calibrating d into a per-direction drivability threshold is a result of chapter 5.

Stop-gate

The difficulty that scales the stop reward is computed over only the obstacles that fall inside the observable window ahead of the vehicle, and it is supplied to the policy as an observation channel, so the stop decision is a causal function of what the agent can currently see rather than of a hidden per-episode label. Keying the stop reward on this same windowed difficulty makes the value of a stop a deterministic function of the observation, which is what lets the

policy learn a clean stop decision. The difficulty channel sets the prize for a correct stop, while perception sets its timing and the progress term credits driving through as much of the layout as possible first. The reverse task needs a further refinement, a state-dependent version of this windowed difficulty that rises as the trailer tracks off the avoidance path, which is developed with its results in chapter 5.

Branching hidden-path reward

Rewarding agreement with a single prescribed avoidance corridor would penalize the agent for choosing an equally valid other side. The branching reward of eq. (4.22) removes this by crediting the nearer of two hidden corridors, so the side of passage becomes an emergent decision from perception. This ties the obstacle task to the observation-space study. Choosing the feasible side and knowing when to stop both require seeing far enough ahead.

4.4 Hyperparameter Optimization

The TD3 agent requires careful hyperparameter tuning. The actor-critic learning rates, replay-buffer size, target-network update rate, and exploration-noise scale are selected over repeated simulation rollouts. TD3 optimization uses step-based updates so that the replay buffer and gradient schedule remain consistent across single-environment and vectorized runs. The throughput of vectorized and GPU-batched execution is addressed in section 4.9. The same backbone and hyperparameters are shared across the state, LiDAR, and BEV observation variants wherever possible, so that the observation ablation isolates the effect of the observation and representation rather than of the training configuration. This sharing cannot extend to the algorithm comparison of chapter 5, where TD3, PPO, and DQN each require the settings appropriate to their own update rule (for example, on-policy batch and rollout lengths for PPO, or the exploration schedule and discretization for DQN). There, only the environment, reward, and observation are held fixed, and each algorithm is given its own sensible configuration so that the comparison reflects the algorithms rather than a single tuning that happens to favor one of them.

4.5 Perception and Representation Design

The core learning system above operates on the engineered state vector. This section introduces the two higher-dimensional observation modalities that this thesis compares against that baseline, a planar LiDAR scan and a bird's-eye-view image, and the representation learning required to turn raw imagery into features the policy can use.

4.5.1 LiDAR Observation

The obstacle environments expose a simulated planar LiDAR observation by appending normalized obstacle distances to the state vector. In forward driving the LiDAR pose is attached to the front tractor, while in reverse driving it is attached to the trailer and rotated by π so that the sensing direction follows the leading unit and covers the area that the system is driving towards. This design keeps the sensing model consistent with the physical task. The front of the articulated system changes when the direction of travel reverses. The LiDAR is a qualitatively different modality from the image, a sparse one-dimensional range scan rather than a two-dimensional picture, and the beam count is treated as a resolution knob in the ablation of chapter 5.

4.5.2 Bird's-Eye-View Observation

The bird's-eye-view (BEV) observation is a 32×32 single-channel top-down image of a local region (approximately a 20 m square) centered on the vehicle and oriented so that the direction of travel points toward the top of the image. It encodes the task-relevant spatial structure that the state vector does not represent: lane geometry, and obstacle layout. Forward and reverse variants share the same size and format and differ only in the crop direction (the forward observation biases the crop toward the upcoming path, the reverse observation toward the region behind the vehicle) and in the interpretation of the accompanying state vector, which uses trailer-referenced tracking errors in reverse because the trailer leads the maneuver.

As introduced in section 3.4, the BEV image originates from the Pygame renderer, which draws the top-down scene as a grayscale frame that is cropped to the local window. For large-scale training the same observation is produced without the renderer by the batched GPU simulator (section 4.9), which rasterizes it analytically as an occupancy image, where each pixel records whether the corresponding ground point is drivable (inside the lane corridor and clear of obstacles) or blocked. The two share resolution and role. The analytic form omits photometric detail and does not draw the vehicle body, which sits at the image center by construction.

Graded (signed-distance) encoding

A binary occupancy image is a poor learning target even though it is a faithful map. It is flat almost everywhere, so the task-relevant quantity, the clearance between the vehicle and the nearest boundary, is carried only by sparse gradients at the cell edges. A randomly initialized convolutional encoder reads those edges as high-frequency noise, which is one route

by which raw vision destabilizes early learning (section 4.6). The BEV channel is therefore optionally rendered as a graded signed-distance field rather than a hard mask. The pixel value rises to +1 at the corridor centerline, falls smoothly to 0 at the drivable edge, and goes negative outside, so clearance becomes a dense, task-aligned signal available at every pixel. Two further coordinate channels in the style of CoordConv [75] can be appended so the convolution can locate a feature within the crop rather than only detect its presence. This graded representation is the image encoding carried forward into the perception ablation of chapter 5.

4.5.3 Image Encoding and Representation Learning

The BEV image must be turned into a feature vector before it can drive the policy. Two routes are studied: training an image encoder end-to-end with the agent, and pretraining an encoder so that a usable visual representation is available from the first control step.

A Convolutional Neural Network (CNN) [43] serves as the end-to-end image front-end, implemented as a custom feature extractor within Stable Baselines3 [72]. Matching the scale of the 32×32 observation, the image branch is deliberately small, a two-layer convolutional encoder with $1 \to 5 \to 5$ channels maps the image to an 80-dimensional feature vector, followed by a linear projection to 64 image features. A separate two-layer multilayer perceptron embeds the 8-dimensional state vector to 64 dimensions. The two branches are concatenated into a 128-dimensional feature vector for the TD3 actor and critic heads. Two choices in this extractor are made specifically for stability with a high-dimensional input: layer normalization is applied to the image features so their scale cannot swamp the intrinsic state embedding, and the actor and critic feature extractors are kept separate, following the TD3 default, so that critic representation updates do not overwrite the actor's features.

Training this image branch end-to-end is not free of pitfalls. Inserting a randomly initialized CNN branch into an otherwise successful state-based policy can harm early TD3 learning. The agent converged to a short-horizon wall-collision behavior even though the same vector state was sufficient for rapid learning. Two diagnostic encoders isolate this effect. The `state_only` BEV encoder keeps the dictionary observation and multi-input policy path but ignores the image branch, confirming that the BEV environment and multi-input wrapper can still learn from the vector state alone. The `scaled_cnn` encoder introduces the image branch gradually through a scalar schedule:

$$\phi(s) = [\lambda_I(k)\, \phi_I(I),\ \phi_x(\tilde{x})]\,, \tag{4.23}$$

where ϕ_I is the image encoder, ϕ_x is the vector-state encoder, and $\lambda_I(k)$ ramps from zero to

one over training. A second schedule can reduce the extrinsic vector components after the image branch has reached full weight:

$$\tilde{x} = [\delta,\ \gamma,\ \lambda_x(k)e_y,\ \lambda_x(k)e_\psi,\ \lambda_x(k)e_{y,t},\ \lambda_x(k)e_{\psi,t},\ \lambda_x(k)\kappa_1,\ \lambda_x(k)\kappa_2]. \tag{4.24}$$

These diagnostics motivate decoupling perception from control, and this thesis pursues a progression of ways to do so, developed in section 4.6. The first and most direct decouples at the level of the representation. If a random image branch can derail learning that the vector state alone proved learnable, then supplying a usable visual representation from the first training step should remove that failure mode. An autoencoder is used to pretrain an image representation before policy optimization [50]. The encoder $E(\cdot)$ maps the image observation s to a feature vector z, and the decoder $D(\cdot)$ reconstructs the image:

$$z = E(s) \tag{4.25}$$

$$\hat{s} = D(z) \tag{4.26}$$

The autoencoder is pretrained on BEV images collected from a mixture of simple pure-pursuit rollouts, already-trained compatible policies, and random actions, to broaden state coverage. After reconstruction training, the encoder weights are transferred into the RL feature extractor and can be frozen or fine-tuned while the policy is trained.

A UNet-style [53] autoencoder provides a second encoder pre-training method with a stronger structural prior. Its encoder uses repeated double-convolution blocks with max pooling, followed by global average pooling to produce a compact 128-dimensional image descriptor. During pretraining the full UNet decoder retains skip connections to improve reconstruction quality. For RL, only the encoder path is kept and concatenated with the vector embedding. It tests whether preserving multi-scale spatial detail during representation learning improves downstream control, particularly in scenes where obstacle shape and lane boundaries must both be interpreted from the image.

Together, the end-to-end CNN, the frozen and fine-tuned autoencoder encoders, and the UNet encoder form a spectrum of image representations, from learned-during-control to learned-before-control, that is compared against the engineered state and LiDAR in chapter 5. Pretraining the encoder is only the first of the decoupling strategies, and on its own it addresses only part of the problem. The stronger routes decouple at the level of the policy and of perception itself, and they rest on a specific view of why raw vision destabilizes learning. That view, and the training methodology built on it, are the subject of the next section.

4.5.4 Observation Space Summary

The full observation interface used across experiments is summarized in table 4.2.

Table 4.2: Observation space components

Component	Description	Role
`vector`	8-dimensional state: $[\delta, \gamma, e_y, e_\psi, e_{y,t}, e_{\psi,t}, \kappa_1, \kappa_2]$	Provides precise articulation and path-tracking state information
`lidar`	N normalized obstacle ranges, with $N \in \{8, 24, 64\}$	Encodes local free space around the leading unit without image processing
`image`	32×32 BEV crop, binary occupancy or graded signed-distance, with optional coordinate channels	Encodes lane layout, local path shape, obstacle positions, and vehicle geometry

4.6 Training High-Dimensional-Observation Policies

Introducing a raw image into the observation does more than add parameters. It changes what it takes to train the policy at all. The diagnostics of section 4.5 showed that inserting a randomly initialized image branch into an otherwise successful state policy can collapse learning that the vector state alone makes easy. This section first sets out a hypothesis for why that happens, then works through three progressively cleaner ways to decouple perception from control: pretraining the representation, warm-starting the policy by imitation, and distilling the state into a frozen perception module. Each route addresses more of the hypothesized failure than the last, and the third recovers ordinary from-scratch learning entirely. The hypothesis is stated here as motivation. The evidence that isolates the cause and separates the three routes is presented in chapter 5.

4.6.1 An Off-Policy Instability Hypothesis

The failure is hypothesized to be an optimization pathology of off-policy actor-critic learning rather than a limitation of perception. In TD3 the actor is trained to maximize the critic's value estimate, while the critic is trained by bootstrapping from its own estimates of the actor's next actions. Early in training the critic is essentially random, and it over-estimates the

value of actions far from the data it has seen. This is extrapolation error [76], a manifestation of the deadly triad that arises when bootstrapping, function approximation, and off-policy updates are combined [20]. With a low-dimensional state the actor recovers quickly. With a high-dimensional image, the actor and its convolutional encoder have far more capacity to chase the critic into these over-valued, off-distribution actions before the critic corrects, so the policy can settle into a degenerate short-horizon behavior. Two ideas from the literature target the same mechanism and inform the designs below: constraining the actor to stay near the behavior data, as in the TD3+BC (Behavior Cloning) offline method [77], and decoupling the encoder from the reward-driven critic gradient, as in the stop-gradient image encoder of SAC-AE [78]. The stronger claim, that this mechanism and not the image content, the representation, or the encoder is the true cause, is tested directly by the causal ablation of chapter 5.

4.6.2 Route A: Pretraining the Representation

The first and most direct response, introduced in section 4.5, is to pretrain the image encoder by reconstruction (the autoencoder and UNet) so that a usable representation exists from the first control step. This attacks the noisy co-trained encoder head-on, but it addresses only one of the two failure sources: it does nothing about the actor being dragged off-distribution by a cold critic, and reconstruction spends encoder capacity on task-irrelevant pixel detail rather than on the geometry the controller needs. It is a partial fix, and it motivates the two stronger routes below.

4.6.3 Route B: Imitation Warm-Start

If a cold critic dragging the actor off-distribution is the problem, then starting from a policy that is already good, and keeping it there, should avoid it. This second decoupling strategy acts at the level of the policy rather than the representation. A competent low-dimensional `state` observation agent is trained first and then used as an expert. It is rolled out inside the image environment, its observations and actions are recorded, and the image policy (convolutional encoder and actor head together) is trained by supervised regression to reproduce the expert's actions. This behavior-cloning warm-start [46] supplies a non-degenerate policy and encoder from the first reinforcement-learning step, so the actor never has to survive a cold critic from scratch. One caveat, revisited in chapter 5, is that the observation still contains the full state vector and the expert's actions derive from it, so the cloning loss can be satisfied largely through the state branch. The warm-start therefore yields a competent, non-collapsing policy, but it does not by itself prove that the convolutional encoder has

learned strong image features. Whether the image supplies information the state lacks is a separate question, one that only becomes measurable on tasks where the state is genuinely insufficient, and is left to future work.

4.6.4 Route C: Geometry Distillation

The cleanest route removes both failure sources at once, by refusing to let reinforcement learning shape the encoder at all. A convolutional encoder is trained supervised to predict the vehicle's geometry directly from the bird's-eye-view image, using the simulator's ground-truth state as a privileged teacher label. This is the privileged-distillation, or learning-by-cheating, paradigm [79]: a teacher that sees the true state supervises a student that must infer it from the image. The trained encoder is then frozen and used as a fixed perception module, and the policy is trained from scratch with ordinary TD3 over a low-dimensional vector, exactly as in the engineered-state baseline. Figure 4.2 shows the two-stage scheme.

The encoder takes the graded signed-distance BEV with its two coordinate channels (section 4.5) through a small three-layer convolutional tower into a 128-dimensional trunk with layer normalization, from which a linear head regresses the standardized geometry vector. The observation the policy receives is then split along the physics of the problem. The internal states, the steering angle δ and the hitch articulation angle γ, are supplied as true proprioception, because they are cheap direct measurements on the vehicle and, being internal, are less reliably recoverable from an ego-centric top-down image, as noted in section 4.5. The exteroceptive geometry, the tractor and trailer tracking errors and the path-curvature look-aheads, is taken from the frozen encoder's prediction, because that is precisely what an external top-down view encodes.

This design removes both failure sources identified above. Because the encoder is trained only by supervised regression and then frozen, the critic is never fed the non-stationary, high-variance features of a co-trained encoder, so the concern behind Route A does not arise. Because the policy is trained from scratch on the frozen features, its actor and critic co-evolve from a cold start, so there is no pretrained-actor-versus-fresh-critic mismatch of the kind Route B must manage. The learning regime is therefore the same as the engineered-state baseline that learns readily. Distillation also settles a question the imitation route could not: predicting the state from the image alone cannot be satisfied through a state branch, so success at the prediction task is direct evidence that the image is a sufficient statistic for the geometry it encodes. The per-component recoverability of the geometry and the from-scratch learning curve are reported in chapter 5.

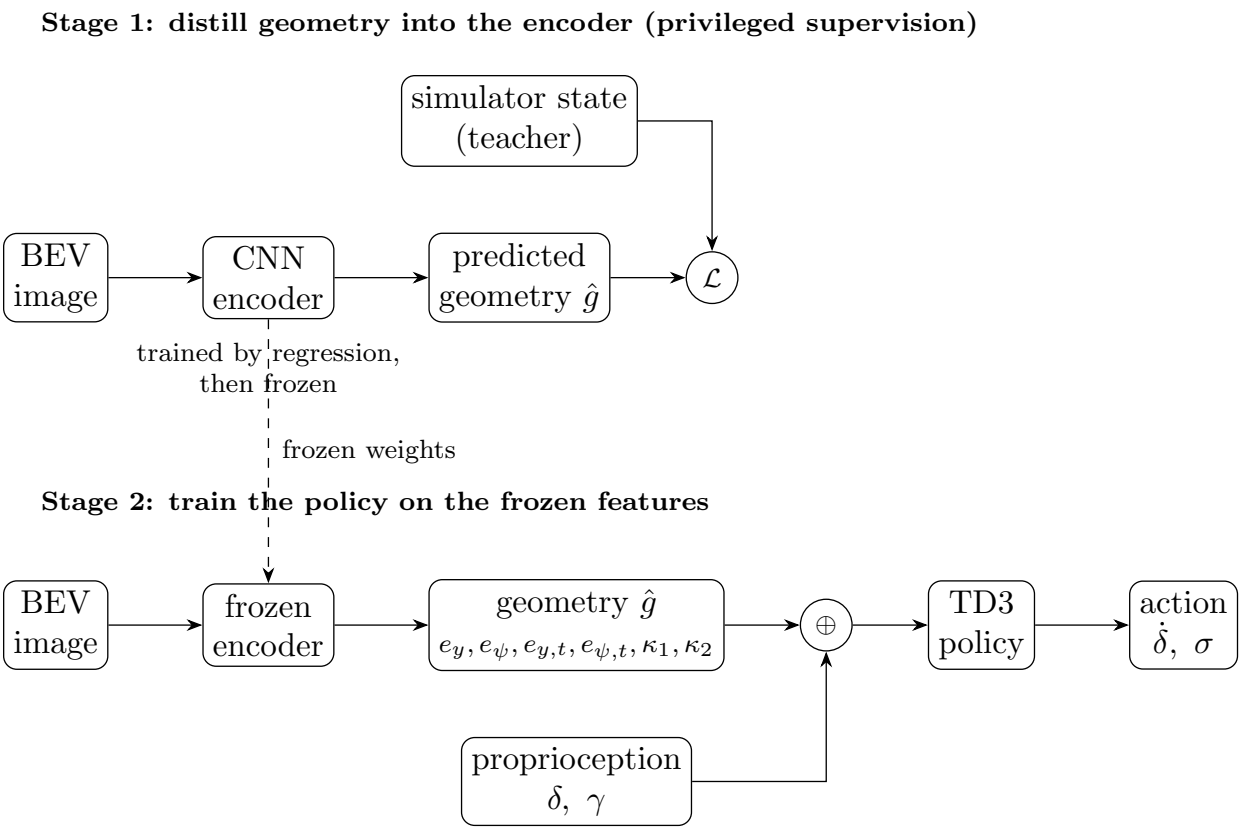


Figure 4.2: Geometry-distillation training. In Stage 1 a convolutional encoder is trained by supervised regression to predict the vehicle's path geometry from the bird's-eye-view image, using the simulator's true state as a privileged teacher, and is then frozen. In Stage 2 the policy is trained from scratch with TD3 on the frozen geometry prediction concatenated with the steering and hitch angles supplied as proprioception.

4.6.5 Safe Adaptation

A warm-started policy raises a further question: can reinforcement learning then improve it without re-triggering the collapse? This is the problem of safe adaptation, and this thesis treats it as a design space rather than a solved method. The most conservative option is to freeze the actor and its encoder and train only the critic, which cannot destabilize a policy it never updates, but for the same reason cannot improve it either. Allowing the actor to move again is where the risk returns. Two anti-collapse designs are identified for this purpose: an action anchor that penalizes departures from the cloned policy, in the spirit of TD3+BC [77], and a frozen-base-plus-residual policy that constrains updates to a small correction around the warm-started actor. Both aim to keep the actor inside the on-distribution region where the critic's values are trustworthy. Their effectiveness, and the prior question of whether the lane-tracking task even leaves reward headroom for adaptation to capture, are examined in chapter 5. The reverse direction, which is both harder to clone and more collapse-prone, is

left as a known gap.

A single annealing schedule underlies these transitions from imitation to improvement: $\alpha_k = \max(0,\, 1 - k/T)$, which decays from one to zero over T steps. It appeared already in the guided reward defined earlier in this chapter, where it blends the pure-pursuit imitation term into the multiplicative task reward, so that the agent first clones the classical controller and is then progressively rewarded for surpassing it. The same idea governs the fine-tuning of an imitated policy. Holding α_k near one keeps the agent close to its imitation while the critic warms up, and decaying it hands control to the task reward once the values are trustworthy. Fixing $\alpha_k = 1$ throughout instead recovers pure imitation, which is how the guided reward is used when a stable clone, rather than improvement, is the goal.

4.6.6 Curriculum and Training Protocol

Two further training choices apply across the obstacle experiments. First, obstacle policies are not trained from scratch but through a two-stage curriculum: Stage A trains a lane-follower in the stop-capable action space with no obstacles, and Stage B warm-starts from it with obstacles enabled, so the agent begins obstacle training already able to track a lane and operate the stop signal. The reverse Stage B is itself warm-started from a pre-validated reverse lane-follower rather than trained from scratch, because reverse learning from scratch is high-variance (chapter 5). This curriculum is distinct from the failure-replay curriculum of section 4.7, which reshapes scenario sampling within a single training run. Second, the off-policy runs use a learning rate of 10^{-3}, since raising it to 2×10^{-3} diverges once the stop action is present, together with 32 parallel environments, the regime found stable for TD3 on this task. On-policy algorithms, which consume far larger environment batches, are configured separately with the per-algorithm settings discussed in section 4.4 and enter only the algorithm comparison of chapter 5.

4.7 Failure Replay Curriculum

A persistent challenge in training RL agents on the reverse lane-following task is that the episode failure rate is high early in training and the lane geometry is re-randomized at every reset. In the standard random-reset regime, each failure is discarded and replaced by a fresh, randomly drawn layout. As a result, the agent rarely encounters the same hard configuration twice, reducing the pressure to master any particular geometry and potentially slowing convergence on the inherently open-loop-unstable reverse task.

To address this, a failure replay curriculum is implemented as a transparent environ-

ment wrapper. At each episode boundary the wrapper inspects the episode outcome. If the episode terminated due to a failure (jackknife, collision, or out-of-bounds departure before reaching the goal), the next reset reuses the same random seed that generated the current path and spawn position, exactly replaying the same scenario. If the episode was completed successfully or timed out, a fresh seed is drawn and a new random layout is used. A configurable consecutive-retry ceiling (default five retries) prevents the agent from being trapped indefinitely on a pathologically difficult layout.

This strategy is a form of automatic curriculum learning [37]: rather than constructing a manually graded task sequence, the curriculum emerges from the agent's own failure signal. Its emphasis on repeatedly revisiting the transitions the agent handles worst is shared, in spirit, with prioritized experience replay [39], though here the emphasis acts at the level of episode sampling rather than replay-buffer weighting. It is implemented orthogonally to the reward formulation and observation representation ablations. The wrapper modifies only the episode-level sampling policy, not the per-step reward signal or the information available to the agent, and is therefore controlled by a separate training flag. Models trained with and without failure replay are saved to distinct directories to ensure clean comparison.

The curriculum is most meaningful for scenarios where the per-episode failure rate is high and layout difficulty is non-uniform, namely reverse lane-following and reverse obstacle avoidance. For forward lane-following the failure rate is low after minimal training and the benefit is expected to be negligible.

4.8 Evaluation Methodology

The observation, reward, and training designs above are assessed by a common evaluation protocol, described once here and applied throughout chapter 5. Its purpose is to make the comparisons fair and the reliability of each result explicit.

Every policy is evaluated on a fixed pool of pre-generated scenarios rather than on freshly sampled episodes, so that different controllers face identical layouts. Outcomes are recorded categorically as completion, deliberate stop, or collision. Because the obstacle task's difficulty distribution is bimodal, results are reported stratified by the geometric difficulty of eq. (4.19) rather than pooled, so that easy and hard layouts are not averaged into a single misleading number. Each reported rate is a proportion, so it is given with a Wilson 95% confidence interval, and each configuration is run over multiple random seeds and reported as a mean with its across-seed interval, so that a result's stability, and not only its central value, is visible.

A raw collision rate is misleading for a policy that can decline to attempt a layout, because

stopping deflates it: a policy that stops on everything never crashes. The honest avoidance metric is therefore the collision rate conditional on attempting, crash/(complete + crash), reported alongside the raw collision rate and the stop rate. The minimum clearance to any obstacle over an episode is also reported, as a measure of how close the vehicle came to contact.

To interpret the stop-gate behavior, the geometric difficulty is calibrated against actual drivability using a classical stack as an oracle. The traditional pipeline (a global path, an artificial-potential-field local planner, and a pure-pursuit tracker, the last of which is one of the classical baselines detailed in appendix A) is run with the stop action disabled so that it always attempts, and its per-difficulty completion rate is read as an empirical drivability $P_{\text{clear}}(d)$. This turns the static geometric proxy of eq. (4.19) into a measured, direction-specific quantity, and it fixes the difficulty at which stopping becomes rational. With the terminal rewards used in this comparison (success +100, collision −500), the expected value of attempting a layout is

$$\mathbb{E}[\text{attempt}](d) = 100\, P_{\text{clear}}(d) - 500\,\big(1 - P_{\text{clear}}(d)\big), \tag{4.27}$$

which is zero at $P_{\text{clear}} = 5/6 \approx 83.3\%$. The difficulty at which measured drivability crosses this break-even is the true, per-direction drivability threshold $d^\star$ against which the stop-gate policy should be judged, rather than the raw geometric scale. The forward and reverse values of $d^\star$, and what they reveal about the reverse task in particular, are reported in chapter 5. So that completion measures tracking rather than stop-timing, the stop action is disabled during the lane-following and drivability evaluations and enabled only when the stop-gate behavior itself is under test.

4.9 Scalable Simulation: GPU-Parallel Batched Environments

The observation, reward, and curriculum studies described in this chapter, together with the controller benchmark of chapter 5, require training and evaluating a large matrix of policies: several observation modalities, multiple reward formulations, forward and reverse variants, and repeated random seeds for statistical validity. The scalar Pygame/Gymnasium environment of section 3.4 is the physically faithful reference model, but it advances a single environment at a time in Python and rebuilds an occupancy grid on every reset, which makes a study of this size prohibitively slow. To make the empirical campaign tractable, the environment was re-implemented as a batched simulator that advances N independent

environments simultaneously on a graphics processing unit (GPU). The original scalar environment is retained unchanged as the parity oracle against which the batched implementation is validated, so that the physical fidelity established in chapter 3 continues to hold for every result reported in this thesis.

4.9.1 Batched Environment Design

The batched environment adds a leading batch axis of size N to every state array, so that a single vectorized operation advances all N environments at once. It is written against a thin array-backend abstraction and runs identically on NumPy for the CPU [80] or CuPy for the GPU [81], selected at run time by a single environment variable. Because CuPy exposes a NumPy-compatible interface, the same vectorized code executes on either device without modification. Achieving this required replacing five constructs in the scalar environment that are inherently sequential or defined one environment at a time:

1. the per-step zero-order-hold discretization, computed in the scalar model with `scipy.signal.cont2discrete`, is replaced by a batched matrix exponential using the scaling-and-squaring method [82], which discretizes all N state-space models in one operation;

2. the per-reset occupancy grid and k-d tree nearest-neighbor lookup are replaced by an analytic distance-to-centerline computation together with an on-device pool of pre-generated paths, which removes the reset bottleneck entirely;

3. the sequential, per-beam LiDAR ray-march is replaced by a vectorized fixed-length march with a first-hit reduction;

4. the Python-loop collision and proximity tests are replaced by a batched gather of vehicle body points against the corridor;

5. the scalar termination, stop-signal, and spawn branches are replaced by boolean masks with internal automatic reset, so that finished environments reset, drawing a new path and setting a new vehicle pose, and restart without host-side control flow.

The batched environment also produces the bird's-eye-view image observation without invoking the Pygame renderer. Because a top-down BEV image is a membership query at a grid of sample points (whether each point lies inside the lane corridor, on an obstacle, or outside the world), it is generated using the same analytic corridor primitives as the LiDAR march: for each environment an $S \times S$ grid is laid out in the vehicle frame, transformed

to world coordinates by the anchor pose, and evaluated at all $N \cdot S \cdot S$ points in a single vectorized gather. The resulting observation is an occupancy image (drivable versus blocked) rather than a rendered grayscale scene, and its cost per environment is comparable to a single LiDAR frame. All three observation modalities (state, LiDAR, and BEV image) therefore run on the GPU, so the entire ablation matrix can be executed on the batched environment.

4.9.2 Numerical Parity

Retaining the scalar environment as an oracle allows the batched implementation to be validated for numerical equivalence rather than merely for plausible behavior. Across a suite of parity tests, the batched dynamics, geometry, observation, and reward pipeline reproduces the scalar model to within floating-point tolerance: the matrix-exponential discretization matches `scipy` to roughly 10^{-8} absolute error, the vehicle dynamics agree to within $\sim 10^{-14}$ over a 200-step rollout, the state and LiDAR observation vector is bit-identical, and the reward matches to $\sim 10^{-14}$ wherever the proximity term is inactive. The NumPy and CuPy backends produce identical results, confirming that the GPU port introduces no numerical divergence of its own.

Two terms are deliberately not bit-identical, because they replace a rasterized grid with analytic geometry: the near-boundary proximity penalty differs by 0.1–0.5 only within the narrow lane-edge region, and the analytic LiDAR march agrees with the grid raycast to within approximately one grid cell, with collision decisions matching the grid on 97–98.5% of randomly sampled states (the remainder lying within one cell of the boundary). These differences are confined to the immediate vicinity of boundaries and do not affect the terminal outcomes that drive reward and evaluation, so the batched environment is a faithful stand-in for the scalar reference in every experiment reported here.

4.9.3 Throughput and Scaling

The batched environment was benchmarked on an NVIDIA RTX 4080 SUPER GPU. Table 4.3 reports raw environment-stepping throughput: at large batch sizes the GPU environment sustains roughly 1.7 million transitions per second for the state observation, against about 2,100 for the single-core scalar environment, and roughly 64,000 transitions per second for the compute-heavier 24-beam LiDAR observation, against about 470. These figures are governed by the batch size and observation type rather than by the vehicle parameters, so they are unaffected by the specific plant used in a given experiment.

Table 4.3: Environment-stepping throughput (transitions per second) for the batched GPU environment versus the single-core scalar reference, measured on an RTX 4080 SUPER.

Observation	Batched GPU (best N)	Scalar (single core)
State	∼1,700,000	∼2,100
LiDAR (24 beams)	∼64,000	∼470

Because the identical code runs on either backend, the GPU's contribution can be isolated by swapping only the array library. Table 4.4 shows this crossover: at small N the GPU's per-kernel launch overhead loses to CPU NumPy, but as N grows the GPU pulls ahead, reaching 6.6× at $N = 4096$ for the state observation. For the compute-bound LiDAR observation the GPU wins at every batch size, up to 62× at $N = 4096$, because each step performs far more work per environment. This crossover is the expected profile of a batched accelerator and motivates the use of large batch sizes ($N \geq 1024$) together with the LiDAR observation to exploit the device.

Table 4.4: Same batched code, NumPy (CPU) versus CuPy (GPU), by batch size N and observation type. Throughput in transitions per second. Speedup is CuPy relative to NumPy.

Observation	N	NumPy (CPU)	CuPy (GPU)	Speedup
State	256	87,060	30,821	0.35×
State	1024	81,668	122,464	1.5×
State	4096	72,200	479,141	6.6×
LiDAR-24	256	2,676	24,815	9.3×
LiDAR-24	1024	2,334	71,394	30.6×
LiDAR-24	4096	1,005	62,601	62.3×

These gains apply to environment stepping, not to the entire training loop. For off-policy algorithms such as TD3, the gradient-update loop remains the bottleneck: it is a sequence of comparatively small-batch optimizer steps that leaves the GPU underutilized (around 34% utilization during off-policy reverse training), so a single off-policy run is not accelerated in proportion to the environment speedup. The batched environment's value for off-policy learning is therefore twofold: it allows many configuration and seed jobs to share one GPU concurrently, and it makes on-policy algorithms, which consume large environment batches directly and so exploit the fast environment, practical to include in the algorithm comparison of chapter 5. To keep that comparison fair, the batched environment is exposed to Stable Baselines3 [72] through a vectorized-environment adapter, so that the vetted imple-

mentations of TD3, SAC, PPO, and DQN run on it unchanged. A separate hand-written GPU-resident TD3 is used only as a throughput demonstration. This infrastructure is what makes the multi-seed ablation campaign of chapter 5 feasible within a practical time budget.

Chapter 5

Results and Discussion

This chapter reports what the learning framework of chapter 4 produced. It follows the same order the design took. The algorithm and the reward are settled first on the engineered state, where learning is fast and the outcome is unambiguous. The observation is then opened up, first to LiDAR and then to the raw image, and the study turns to what richer sensing buys and what it costs. Making the image usable at all turns out to be a training problem rather than a perception problem, and isolating that problem is the central result of the chapter. The failure-replay curriculum, the obstacle-navigation task, and the comparison against tuned classical controllers close the chapter. Throughout, the aim is to separate two questions that are easy to conflate, whether the articulated control task benefits from richer sensing, and whether any benefit comes from the sensing itself or from the machinery used to turn it into features.

5.1 Evaluation Setup

Every number in this chapter comes from the common evaluation protocol described once in section 4.8, so only the points that bear on reading the tables are repeated here. Policies are scored on a fixed pool of pre-generated scenarios rather than on freshly sampled episodes, so that different controllers face identical layouts. Each configuration is trained over three random seeds and reported as a mean with a 95% confidence interval, so that the stability of a result, and not only its central value, is visible. This matters here because articulated control training is high-variance, and on the harder tasks the spread across seeds carries as much of the story as the mean does. The lane-following results use best-checkpoint completion with the stop action disabled, so that completion measures tracking quality rather than a policy's willingness to decline a layout. The obstacle results, where declining to attempt is itself part of the behavior, use the attempt-conditional metrics of section 4.8 instead.

Generalization is treated as something to measure rather than assume. The evaluation pool is held out from training, and where two methods are close in mean performance the across-seed interval, rather than the point estimate alone, decides whether the difference is real. A gap of a few centimeters in mean cross-track error routinely sits inside the run-to-run scatter on the reverse task, and reporting a single best number would read that scatter as an effect.

A word on the evaluation pool is in order before the numbers. The pool is the unfiltered mild-path distribution, every generated layout kept, and every controller is scored on the identical set. An earlier version of the pool kept only paths a tuned pure-pursuit controller could complete, which guaranteed pure pursuit a perfect score by construction and biased any comparison against it. Removing that filter shifts the reverse rates by at most about two points and changes none of the orderings, so the results reported here are on the unfiltered pool throughout. The qualitative findings, the orderings, the negative results, and the perception analysis do not depend on the pool at all.

5.2 Selecting the Algorithm and the Reward

The first question is which learning algorithm and which reward give stable articulated tracking, and the cleanest place to answer it is the engineered state, where perception adds nothing to confound the comparison. Four algorithms, TD3, PPO, a discrete-action PPO, and DQN, were each trained under four reward formulations, in both the forward and the reverse direction, over three seeds. The environment, the observation, and the evaluation were held fixed, and each algorithm was given the settings appropriate to its own update rule as described in section 4.4. Tables 5.1 and 5.2 report best-checkpoint completion for every cell.

Table 5.1: Forward lane-following, best-checkpoint completion rate (%) for every algorithm and reward combination on the engineered state observation. Cells are the mean over three seeds with a 95% confidence interval. The chosen recipe is highlighted.

Algorithm	Dense	Tractor-focus	Multiplicative	Guided (PP)
TD3	100.0±0.0	100.0±0.1	100.0±0.0	100.0±0.0
PPO	100.0±0.0	100.0±0.0	100.0±0.0	100.0±0.0
DQN	100.0±0.0	100.0±0.0	85.0±15.2	60.6±59.1
PPO (discrete)	83.3±32.6	90.7±18.2	100.0±0.0	49.0±1.5

Table 5.2: Reverse lane-following, best-checkpoint completion rate (%) for every algorithm and reward combination on the engineered state observation. Cells are the mean over three seeds with a 95% confidence interval. The chosen recipe is highlighted.

Algorithm	Dense	No-hitch	Multiplicative	Guided (PP)
TD3	79.2±4.3	80.5±12.8	**95.9±0.6**	86.1±6.5
PPO	16.1±31.4	16.4±32.1	48.2±0.8	82.0±24.6
DQN	16.9±32.8	1.4±2.8	0.0±0.0	48.7±55.1
PPO (discrete)	47.5±1.2	32.5±31.9	15.8±31.0	57.3±8.9

Forward lane-following is saturated. Both continuous-control algorithms complete every forward layout under every reward, which matches the classical baselines, where pure pursuit, PID, LQR, and MPC all reach full completion forward. Only the combination of a discrete action space and a poorly matched reward breaks the forward task at all. Forward tracking is therefore not where the algorithm or the reward is decided, and the forward reward choice is left to the tracking-quality detail rather than to completion.

Reverse lane-following is where the choice is made. The open-loop reverse dynamics are unstable, as established in chapter 3, so early policies fail often and the algorithm has to recover a stabilizing controller rather than merely refine a tracker. TD3 is the only algorithm that solves the reverse task, with every one of its reward variants completing at least 79% of the pool. The best non-TD3 reverse cell, PPO under the guided reward at 82.0%, carries a confidence interval of ±24.6, so its apparent success rests on a single lucky seed rather than on reliable learning. The best reward is also algorithm-conditional. TD3 pairs best with the multiplicative reward, which forces the tracking and articulation errors down jointly, whereas the on-policy PPO only works under the guided reward that clones the pure-pursuit teacher, and the multiplicative reward is close to catastrophic for DQN. The guided reward fails both discrete algorithms in both directions, which is the expected behavior of a reward that clones a continuous teacher into an action space that cannot represent it, and serves as a clean negative control.

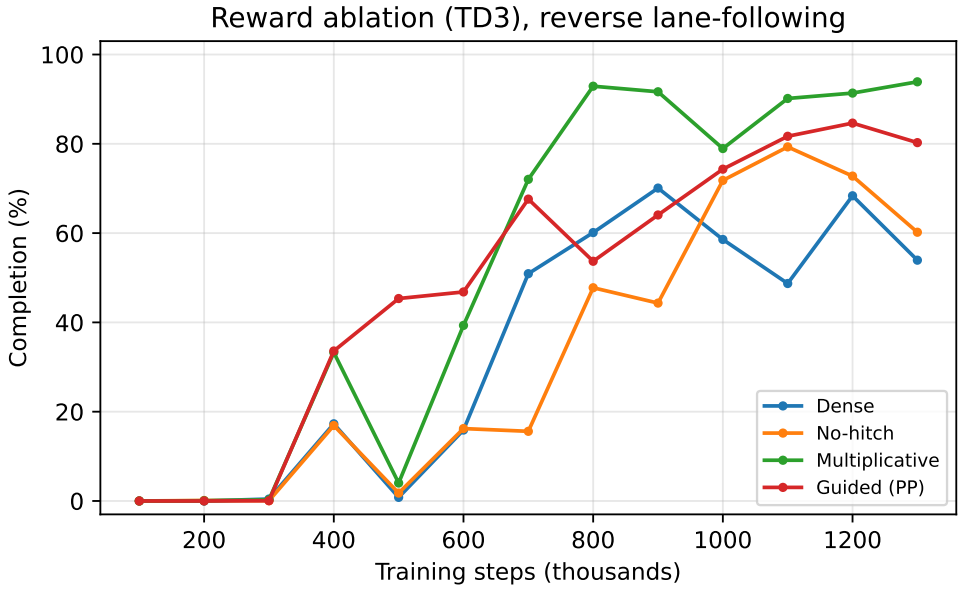


Figure 5.1: Reverse lane-following completion against training steps for TD3 under each reward formulation. The multiplicative reward reaches the highest and steadiest completion.

The chosen recipe is therefore TD3 with the multiplicative reward, and the reverse task is where that choice is justified. Table 5.3 looks past completion to the tracking detail for TD3, and the multiplicative reward is best on every axis at once, the highest completion at 95.9%, the lowest trailer cross-track error, and the tightest hitch behavior. This single configuration, TD3 with the multiplicative reward on the engineered state, is the configuration carried into the rest of the chapter. Its reverse completion of 0.959 ± 0.006 sits about two points under pure pursuit at 0.980, and it does so with no collapsed seeds. Holding this recipe fixed is what lets every later ablation change one design variable at a time.

Table 5.3: Reverse lane-following, TD3 reward detail on the engineered state observation. Beyond completion the multiplicative reward also gives the lowest trailer cross-track error and the tightest hitch behavior, which is why it is carried forward. Mean over three seeds, 95% confidence interval on completion.

Reward	Completion (%)	Trailer CTE (m)	Max $\lvert\gamma\rvert$ (°)	Jackknife (%)
Dense	$79.2_{\pm 4.3}$	1.14	17.5	1.2
No-hitch	$80.5_{\pm 12.8}$	1.01	14.7	0.5
Multiplicative	$\mathbf{95.9_{\pm 0.6}}$	0.94	12.3	0.3
Guided (PP)	$86.1_{\pm 6.5}$	1.20	11.5	2.0

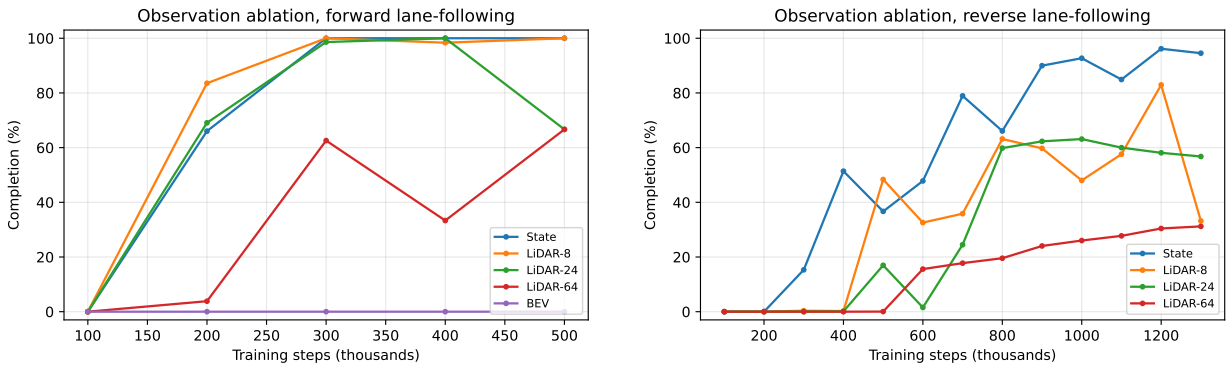
5.3 Observation-Space Ablation

With the algorithm and reward fixed, the observation can be opened up. The engineered state is a compact, hand-designed summary of exactly the quantities a tracking controller needs. LiDAR and the bird's-eye-view image carry more of the raw scene but leave the policy to extract those quantities for itself. The question this section answers is what that trade buys. The comparison uses the identical TD3 recipe for every observation, with no curriculum, no failure replay, and no representation warm-up, so that any difference is attributable to the observation and its dimensionality rather than to the training treatment. Table 5.4 reports completion and trailer cross-track error for the engineered state, three LiDAR resolutions, and the raw image, in both directions.

Table 5.4: Observation-space ablation on no-obstacle lane-following, identical TD3 recipe (multiplicative reward, fixed speed) for every observation. Best-checkpoint completion and trailer cross-track error, mean over three seeds with a 95% confidence interval on completion. Reliability falls as the observation grows, and the effect is sharper in reverse. The reverse BEV run is deferred to the vision-rescue study.

Observation (dim)	Forward completion (%)	Forward CTE (m)	Reverse completion (%)	Reverse CTE (m)
State (8)	$100.0_{\pm 0.0}$	0.34	$97.0_{\pm 0.4}$	1.01
LiDAR-8 (10)	$100.0_{\pm 0.0}$	0.41	$90.0_{\pm 12.5}$	1.31
LiDAR-24 (26)	$100.0_{\pm 0.0}$	0.46	$63.1_{\pm 61.9}$	1.21
LiDAR-64 (66)	$66.7_{\pm 65.3}$	0.95	$31.2_{\pm 61.1}$	1.00
BEV (32×32)	$0.0_{\pm 0.0}$	1.17	—	—

The result contradicts the expectation that more sensing is better. Reliability falls as the observation grows, and the effect sharpens on the harder reverse task. The engineered state is not merely sufficient but the most robust option in both directions. The LiDAR variants show the same pattern internally, which reframes the beam count as a resolution-against-reliability knob rather than a more-is-better axis. The 8-beam scan is the most robust of the LiDAR options, matching the state forward and reaching 90% reverse, while raising the count to 24 and then 64 beams widens the across-seed spread and then brings on outright seed collapses. The wide confidence intervals on the reverse LiDAR-24 and LiDAR-64 rows are not noise to be averaged away, they are the signature of that collapse, with individual seeds falling to zero completion while others still succeed. Because carrying a sensing modality onto hardware is a separate concern taken up in chapter 6, the point retained here is only that added input dimensions cost reliability on this task rather than earning it.



(a) Forward

(b) Reverse

Figure 5.2: Observation-space ablation learning curves, completion against training steps for the identical TD3 recipe under each observation. The raw image never leaves zero, and the higher-dimensional LiDAR scans learn less reliably than the compact state.

The raw image is the extreme case. Trained from scratch with the same recipe that solves the state task, the bird's-eye-view policy never completes a single forward layout, and the reverse image run was set aside for the dedicated study of the next section. This is a striking outcome, because the image plainly contains the geometry the state encodes, and more besides. Whatever is stopping the image policy from learning, it is not a shortage of information in the observation. The next section examines why.

5.4 Training Vision-Based Policies

An image observation that contains the exteroceptive geometry the engineered state summarizes, yet cannot learn the task the state solves easily, is a contradiction that must be resolved, because the obstacle task of section 5.5 requires the image. The hypothesis set out in section 4.6 is that the failure is an optimization pathology of off-policy actor-critic learning rather than a limit of perception. A randomly initialized image encoder gives the actor and its convolutional front-end far more capacity to chase a cold, over-optimistic critic into off-distribution actions than the compact state does, and the policy collapses into a degenerate short-horizon behavior before the critic corrects. This section reports the evidence that isolates that cause and then the recipe that removes it.

5.4.1 Isolating the Failure Mode

The evidence takes the form of a ladder, where each rung changes exactly one thing relative to the rung above, so that the first configuration to recover completion identifies what was

actually broken. Table 5.5 lays it out on the forward task.

Table 5.5: The vision-training ladder on forward lane-following. Each rung changes one thing relative to the rung above, so the first configuration that reaches full completion identifies the cause. Representation changes alone leave the policy at zero, imitation restores it, unconstrained fine-tuning destroys it again, and only freezing the actor holds it, which points to the actor rather than the encoder as the failure. Completions are from the development runs. The actor-frozen configuration was repeated over three seeds (1.00 / 1.00 / 0.92), and per-seed archival of the remaining rungs is pending.

Configuration	Completion	What it isolates
BEV occupancy, from scratch	0.00	Ordinary RL on the raw image fails outright
+ signed-distance encoding	0.00	A denser, task-aligned image is not the fix
+ coordinate channels	0.00	Spatial addressing is not the fix
+ stability extractor	0.00	Layer normalization and separate actor and critic encoders are not enough
Imitation warm-start	1.00	A cloned state expert produces a working vision policy
+ naive fine-tuning	0.00	Letting RL move the whole policy destroys the clone
+ RL, encoder frozen	0.00	Freezing perception does not save it, so the encoder is not the culprit
+ RL, actor frozen	1.00	Freezing the actor and training only the critic holds parity

The representation-side fixes come first and none of them work. Rendering the image as a graded signed-distance field rather than a hard occupancy mask, adding coordinate channels, and giving the encoder the stabilizing treatment of layer normalization with separate actor and critic front-ends all leave completion at zero. If the trouble were that the image is a poor learning target, one of these would have improved completion. None does, which points away from the representation as the cause. The picture changes the moment the policy is warm-started by imitation. Cloning the competent state expert into the image policy produces a working controller with full completion, which shows that a non-degenerate policy over the image is reachable. The decisive rungs are what happens next. Letting ordinary reinforcement learning then fine-tune that cloned policy destroys it, dropping completion back to zero, and freezing the encoder while the rest of the policy trains does not save it

either. Only freezing the actor, and training the critic alone, holds the cloned policy at parity. Read down the ladder, the cause is neither the image nor the encoder but the actor being driven off-distribution by an untrained critic, exactly the mechanism hypothesized in section 4.6. The actor-frozen configuration was repeated over three seeds and held, at 1.00, 1.00, and 0.92 completion.

This diagnosis also exposes why the imitation warm-start, though it works, is not a complete result. Because the image observation still carries the full state vector, the cloning objective can be met largely through the state branch, so a working cloned policy does not on its own prove that the convolutional encoder learned to read the image. Settling that requires an approach where the image alone must supply the geometry.

5.4.2 Distilling Geometry into a Frozen Encoder

The recipe that removes the failure at its root refuses to let reinforcement learning shape the encoder at all, following section 4.6. A convolutional encoder is trained by supervised regression to predict the vehicle and path geometry directly from the bird's-eye-view image, using the simulator's true state as a privileged teacher label. It is then frozen and used as a fixed perception module, and the policy is trained from scratch with ordinary TD3 over a low-dimensional vector, exactly as in the engineered-state baseline. The internal steering and hitch angles are supplied as true proprioception, since they are cheaply available from onboard sensing and state estimation and are not reliably visible in a top-down image, while the exteroceptive path geometry is read from the frozen encoder.

Two measurements test this recipe. The first asks whether the image is a sufficient statistic for the geometry, by scoring how well each state component can be predicted from the image alone. Table 5.6 reports the per-component test R^2. The exteroceptive geometry, the tractor and trailer tracking errors and the two curvature look-aheads, is recovered almost perfectly, with R^2 at or above 0.97 for seven of the eight components. Only the internal steering angle resists, at 0.66, which is exactly the component that a top-down view of the surroundings should not be able to see, and precisely why steering is supplied as proprioception rather than predicted. Because predicting the state from the image alone cannot be satisfied through a state branch, this is the direct evidence the imitation route could not provide, the image really does encode the path geometry.

Table 5.6: Per-component test R^2 of predicting each state variable from the bird's-eye-view image alone, using the frozen geometry encoder. The exteroceptive path geometry is recovered almost perfectly, while the internal steering angle resists, which is why steering and the hitch angle are supplied as true proprioception rather than predicted.

Component	Description	R^2
e_ψ	tractor heading error, exteroceptive	0.998
e_y	tractor cross-track error, exteroceptive	0.997
κ_1	near curvature sample, exteroceptive	0.997
κ_2	far curvature sample, exteroceptive	0.997
γ	hitch angle, partly visible through the path history	0.995
$e_{\psi,t}$	trailer heading error, exteroceptive	0.993
$e_{y,t}$	trailer cross-track error, exteroceptive	0.974
δ	steering angle, an internal actuator state the image cannot see	0.659

The second measurement asks whether a policy can then learn on these frozen features. Figure 5.3 shows the from-scratch learning curve. Completion climbs monotonically to full parity by roughly 300 thousand steps, with no imitation, no critic warm-up, no unfreezing schedule, and no collapse, tracing the same trajectory as the engineered-state run. The image policy that could not learn at all in section 5.3 learns cleanly once the encoder is decoupled from the reward-driven gradient. Cross-track error at parity averages 0.55 m against 0.46 m for the engineered-state policy trained under the same budget, a gap consistent with the few-percent prediction error the encoder carries, though the per-seed spread on both runs is wide enough that the two overlap.

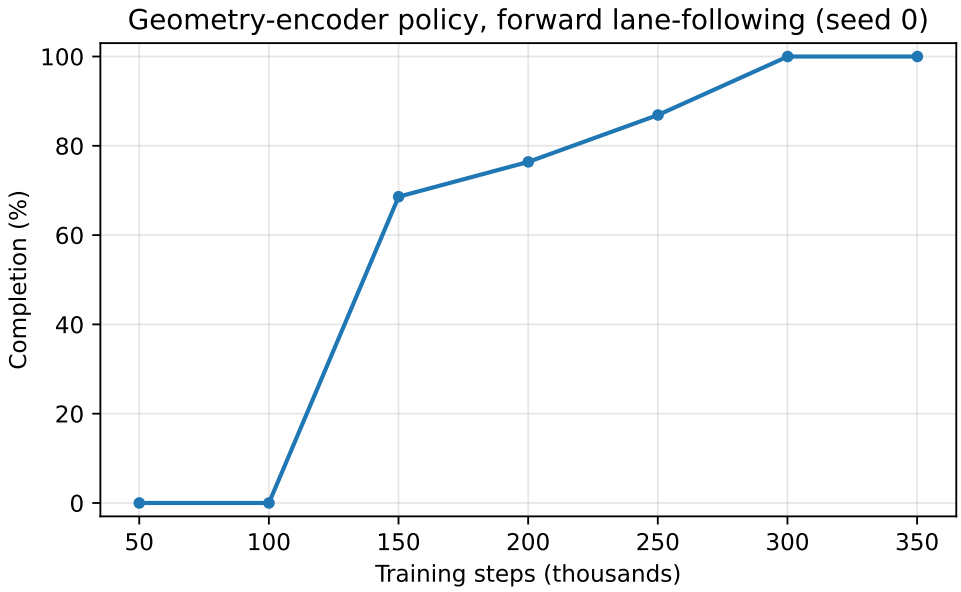


Figure 5.3: Geometry-encoder policy completion against training steps on forward lane-following, trained from scratch on the frozen encoder. Completion reaches parity with the engineered-state baseline by about 300 thousand steps with no collapse.

The single-seed curve is confirmed by the full three-seed set. Forward completion reaches 1.000 on every seed, matching the engineered-state policy trained under the same budget, and the reverse direction reaches 0.954 against that policy's 0.967. Reverse needed a longer budget to get there. At 1.3 million steps it averaged 0.831 with wide seed variance, and extending to two million steps recovered the worst seed from 0.646 to 0.910. The limiter was therefore training convergence rather than the encoder or the replay buffer, which rules out the state-only-replay rework the earlier result appeared to call for.

One limitation remains. The question of whether reinforcement learning can safely improve a warm-started policy without re-triggering the collapse was probed directly. Freezing the actor holds parity but by construction never improves the policy, and allowing the actor to move again, even under a reduced learning rate, either collapsed the policy outright or left it oscillating with no improvement in cross-track error, [pending: safe-adaptation collapse rates and CTE]. Unconstrained actor unfreezing is therefore not yet a usable recipe on this task, and whether the lane-tracking reward even leaves headroom for adaptation to capture is left open.

5.5 Obstacle Navigation

The obstacle task is where the preceding results converge. It is the one setting where the engineered state is genuinely insufficient, because a compact tracking summary is blind to obstacles that are not on the reference path, so the image has to carry information the state cannot. It is also where the value of a learned controller and perception system separates cleanly from classical tracking, because avoiding an obstacle is a joint planning-and-control problem rather than a pure tracking one. The approach taken here is not a monolithic mapping from pixels to actions, which section 5.4 showed does not train. It is a learned but interpretable geometric bottleneck, and the results below build it up one decision at a time.

5.5.1 Direct Obstacle Encoding

The direct approach is to hand the policy the obstacles, as a set of ordered range-and-bearing slots appended to the state. It fails. Trained under the same curriculum, the slot-based policy climbs to a modest completion and then collapses toward zero, and it does so under every slot ordering tried, nearest-first, fixed, and randomized alike. Because a fixed ordering with no reordering collapses just as the others do, the instability is not caused by slots swapping identity between steps. It is the same off-policy training pathology seen in section 5.4, now provoked by a different high-dimensional input. This rules out explicit obstacle encoding and motivates a representation that does not ask the policy to parse a variable set of objects at all.

5.5.2 Observing the Avoidance Path

The approach that succeeds gives the policy a path rather than a scene. The environment already computes a hidden avoidance path, a potential-field offset from the centerline that bends around the obstacles, and the reward already tracks it. Instead of feeding the policy the obstacles, the policy is fed the tracking errors and curvature to this avoidance path. Obstacle avoidance then reduces to path following, the task already solved in the preceding sections, and the obstacle information arrives pre-digested as the path to follow. The image predicts this path, in the same learning-by-cheating manner as the geometry encoder of section 5.4.2, privileged in simulation and image-derived at deployment.

A ground-truth test isolates the mechanism from the perception. Taking an existing forward state policy and feeding it avoidance-path errors in place of centerline errors, with no retraining at all, cut obstacle collisions roughly in half and lifted completion from about 0.55 to about 0.67. The controller follows whatever path it is given, so the entire obstacle

problem can be moved into the choice of path, which is exactly the quantity the image can supply.

5.5.3 The Stop Decision and Forward Results

Following the avoidance path is not enough on its own, because some layouts are not passable and the right action there is to stop rather than to clip an obstacle. The policy is therefore given a windowed difficulty channel, the free-gap difficulty of eq. (4.19) computed over the obstacles inside the observable window, so that the attempt-versus-stop decision is a causal function of what the agent can currently see. As set out in section 4.3.1, keying the stop reward on this same windowed difficulty is what makes the value of a stop predictable from the observation. An earlier design that keyed the reward on a global per-episode difficulty made the stop value unpredictable from the observation and collapsed the policy into stopping everywhere, and sharpening the stop calibration, rather than lowering its magnitude, concentrated stopping on the genuinely blocked layouts.

In the forward direction this static-difficulty recipe works well. Table 5.7 reports the outcome stratified by layout difficulty. The policy drives through easy layouts, stops on every hard one, and does not crash on the hardest, for an overall safe rate of 0.907 at a 0.019 crash rate. The residual crash concentrates in the tight-but-passable band where the attempt-versus-stop call is genuinely ambiguous. This behavior is what the drivability calibration of section 4.8 predicts. Calibrating the geometric difficulty against the classical drivability oracle through the expected-value balance of eq. (4.27) puts the forward break-even near a difficulty of 0.8, so stopping on the hardest layouts is the rational choice rather than a shortfall.

Table 5.7: Forward obstacle maneuvering, retained static-difficulty model, outcomes stratified by layout difficulty. The policy drives through easy layouts and stops on every hard one with no crash on the hardest, for an overall safe rate of 0.907 at a 0.019 crash rate. The residual crash concentrates in the tight-but-passable band.

Difficulty	Complete (%)	Stop (%)	Crash (%)
$[0.2, 0.4)$ easy	83.1	15.5	1.4
$[0.4, 0.6)$	53.0	40.8	6.3
$[0.6, 0.8)$	0.0	100.0	0.0
$[0.8, 1.0)$ blocked	0.0	100.0	0.0

5.5.4 State-Dependent Difficulty in Reverse

Reverse is the hardest case, and the forward recipe does not transfer to it. A static difficulty describes the layout, the free gap, but not the situation, and in reverse the two come apart. Because reverse tracking runs a systematic off-path error of around 0.8 m, tracking well at one moment does not imply the trailer will clear the coming gap, so a policy gated on layout difficulty over-attempts in the ambiguous band and clips obstacles there, crashing on around forty to forty-five percent of the attempts it makes in that band.

The fix is to make the difficulty state-dependent. The effective vehicle width is inflated by the current trailer error to the avoidance path rather than by an average, so when the trailer is tracking off-path into an obstacle the usable gap shrinks and the difficulty rises in time to trigger a stop. A second correction concerns where the difficulty looks. In reverse the trailer leads, so a window that looked only ahead of the lead point was blind to obstacles the trailing body was still clearing, which produced a spurious cluster of crashes at apparently low difficulty. Anchoring the window to the whole vehicle body removed it. This is a reverse-specific need, as the collision geometry in table 5.8 makes plain. Reverse crashes are overwhelmingly trailer clips, the leading trailer sweeping into an obstacle, whereas forward crashes are entirely tractor clips, so the body-extent window matters in reverse and is irrelevant forward.

Table 5.8: Collision geometry by direction. Reverse crashes are dominated by trailer clips, which the body-extent difficulty window addresses, while forward crashes are entirely tractor clips, so the window fix is a reverse-only need.

Direction	Tractor clip (fraction)	Trailer clip (fraction)
Forward	1.00	0.00
Reverse (mean of 3 seeds)	0.18	0.82

With the state-dependent difficulty and the body-extent window in place, the stop decision reduces to a single clean threshold. Measured at the moment of collision, no reverse crash occurs below a dynamic difficulty of about 0.5, so a stop below that value is unambiguously wrong and a stop at or above it is the best available action. Figure 5.4 shows the outcome stratified by dynamic difficulty, and table 5.9 gives the same data numerically. Completion falls and stopping rises across the gate, and the crash rate stays low throughout, with the accepted model reaching an overall crash rate of about 0.05.

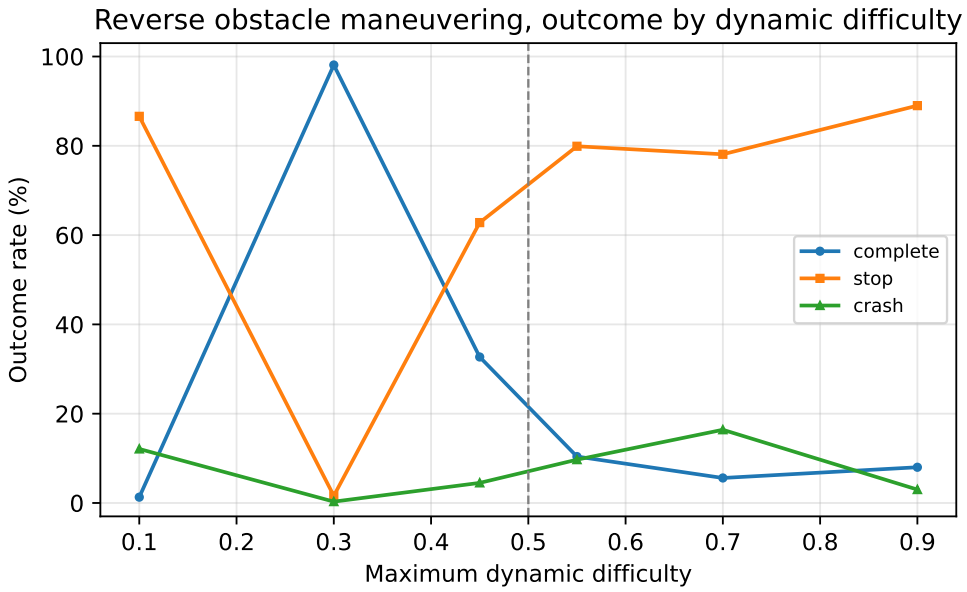


Figure 5.4: Reverse obstacle maneuvering, outcome rates against maximum dynamic difficulty. No crash occurs below the gate near a dynamic difficulty of 0.5, so stopping above it is the correct action.

Table 5.9: Reverse obstacle maneuvering, accepted dynamic-difficulty model, outcomes stratified by the maximum dynamic difficulty reached. Completion falls and stopping rises with difficulty, and no crash occurs below the gate at roughly 0.5. The low-difficulty stops are the accepted articulation limitation, not spurious behavior.

Dynamic difficulty	n	Complete (%)	Stop (%)	Crash (%)	Stop $\lvert\gamma\rvert$ (rad)
$[0.0, 0.2)$	762	1.3	86.6	12.1	0.204
$[0.2, 0.4)$	2239	98.1	1.6	0.3	0.105
$[0.4, 0.5)$	584	32.7	62.8	4.5	0.039
$[0.5, 0.6)$	1160	10.4	79.9	9.7	0.101
$[0.6, 0.8)$	1217	5.6	78.1	16.4	0.119
$[0.8, 1.0)$	5778	8.0	89.0	3.0	0.062

The reverse recipe is trained from scratch and repeated across three seeds in table 5.10. Two of the three reproduce the safe result at a crash rate around 0.05, while one seed converged to a less conservative policy that attempts more of the transition band and crashes more there. The spread traces to a critic shock at the boundary where the reward changes from the obstacle-free warm-up to the obstacle task, so the recipe is usable but not yet tightly

reproducible. A critic warm-up at that boundary and a safety-aware checkpoint criterion are the levers to tighten it, and both are left as immediate follow-ups. The same table records a second finding. Dynamic difficulty helps reverse but hurts forward, where the vehicle does not jackknife and its tracking error is small, so the width inflation only adds noise. Forward therefore keeps the static recipe, and a single unified stop signal for both directions is not achievable given their different dynamics.

Table 5.10: Reverse dynamic-difficulty model across three seeds trained from scratch. Two of three seeds reproduce the safe result, and seed 1 converged to a less conservative policy.

Seed	Complete (%)	Stop (%)	Crash (%)
Seed 0	26.0	68.8	5.2
Seed 1	36.8	49.7	13.5
Seed 2	32.8	62.3	5.0
Crash mean ± sd			$7.9_{\pm 4.9}$

Table 5.11: Forward crash rate, static against dynamic difficulty. Dynamic difficulty hurts forward, so forward keeps the static recipe.

Forward recipe	Crash (%)
Static difficulty	2.7
Dynamic difficulty (mean of 3 seeds)	8.7

5.5.5 Recovering the Difficulty from the Image

The obstacle recipe gates on the dynamic difficulty, which depends on the current trailer error and so is a harder image target than the static path geometry of section 5.4.2. The same distillation recovers it. Table 5.12 reports the per-component recoverability of the avoidance-path state from the obstacle image in both directions, and the dynamic difficulty channel is recovered well, at a test R^2 of 0.948 forward and 0.954 reverse. The avoidance-path errors are recovered strongly, while the avoidance-path curvature is softer than in the lane task and softer still in reverse, where the bends are sharper, though the errors and the difficulty channel carry the control.

Table 5.12: Recoverability of the avoidance-path state from the obstacle image, test R^2 per component for the forward static and reverse dynamic recipes. The dynamic difficulty channel is highly recoverable in both directions, so there is no perception roadblock. Reverse avoidance-path curvature is softer than forward.

Component	Forward R^2	Reverse R^2
e_y	0.957	0.980
e_ψ	0.932	0.943
$e_{y,t}$	0.981	0.977
$e_{\psi,t}$	0.991	0.936
κ_1	0.965	0.726
κ_2	0.903	0.639
d_{win}	0.948	0.959

Closing the loop confirms there is no perception roadblock. Table 5.13 drives each policy on its own image-predicted state, with a frozen encoder in the loop and only the steering and hitch angles taken from proprioception. The full pipeline holds in both directions, with the crash rate sitting a few points above the ground-truth-state policy from ordinary prediction noise, a larger margin in reverse where the curvature is harder to see. The image supplies both the plan, the avoidance path and its difficulty, and the control state, which is the claim the whole perception thread of this chapter was built to support.

Table 5.13: Closed-loop vision pipeline, policy driven on image-predicted avoidance-path state with a frozen encoder in the loop. The full pixels-to-control pipeline holds in both directions, a few points of crash above the ground-truth-state policy from prediction noise.

Recipe	Difficulty R^2	SAFE (%)	Crash (%)
Forward (static)	0.948	90.3	4.1
Reverse (dynamic)	0.954	74.9	8.3

5.5.6 Deterministic Safety Gate

The conservative stops seen above are not spurious behavior. On predominantly straight reverse paths the policy occasionally halts where a continued maneuver would have completed, and at those halts the trailer's hitch articulation is already degrading, with a terminal articulation angle around 0.20 rad against about 0.01 rad on completed paths, trending toward

the unrecoverable regime near 0.71 rad that produces the collisions. A reversing tractor-trailer cannot unwind a growing hitch angle by continuing in reverse, because the reverse articulation dynamics are unstable and more reverse motion only grows the hitch. The geometrically required recovery is to pull forward, re-stabilize the articulation on the stable forward dynamics, and re-approach, which is a shunting maneuver. That recovery needs a forward-reverse switching action space that is outside the scope of this continuous-reverse study, so within a reverse-only action space halting is the correct and safe response, and permitting pull-forward recovery is left to future work.

Rather than rely on the policy erring conservative, the same protection can be made explicit and controller-independent with a deterministic safety gate. The gate stops the vehicle before it commits to a maneuver whenever it is in a bad enough state, using two thresholds for the two ways a reverse maneuver crashes. It halts if the peak dynamic difficulty exceeds 0.375, which catches the obstacle clips where the vehicle tracks well but the gap is too tight, and it halts if the peak hitch angle exceeds 0.6 rad, which catches the articulation runaways. The hitch threshold sits above the 99th percentile of successful maneuvers, about 0.40 rad, and below the articulation-crash floor near 0.71 rad, so it catches every articulation crash while rarely stopping a maneuver that would have completed. Applied after training with no retraining, on the reference seed the gate reduces the reverse crash rate from 0.052 to 0.0004, about five episodes in 11,740.

This gate is what sets the learned and classical controllers apart, and then brings them back together. Without any deterministic gate the learned attempt-versus-abort decision separates the two controllers sharply, because pure pursuit has no way to decline an infeasible layout and crashes on 54 percent of the hard-layout-dominated pool, against about 5 percent for the learned policy. Once the deterministic gate is layered on top, the two converge, with the crash rate driven to near zero for both and comparable completion, as Table 5.14 shows. The learned stop decision is therefore what makes the policy safe when no hand-designed gate is available, and the deterministic gate is an added safety layer that a classical controller can equally use.

Table 5.14: Reverse obstacle maneuvering with and without the deterministic safety gate, learned policy against pure pursuit on the same hard-layout-dominated pool. Without a gate the learned attempt-versus-abort decision already keeps the learned policy far safer than pure pursuit, which cannot decline a layout. Adding the gate drives the crash rate to near zero for both and leaves their completion comparable. Reference seed. Commit-success is completions over completions plus crashes.

Controller	Crash, native stop no gate (%)	Crash, with gate (%)	Completion, with gate (%)	Commit success (%)
Learned (RL)	5.2	0.0	17.7	99.8
Pure pursuit	54.0	0.2	18.1	99.0

Together, these results explain why a learned controller is warranted here even though a classical planner exists. Its value is the learned attempt-versus-abort stop decision, which a geometric planner does not make, together with the reverse articulated case, where a geometric avoidance path is not by itself trackable and the stop must be tied to the articulation state. Both live in one learned module that reads its plan and its control state from the same image, in place of the separate global planner, local planner, and tracker of the classical stack. The deterministic gate does not displace that value, it adds a controller-independent safety floor on top of it. The comparison against that stack is taken up next.

5.6 Comparison with Classical Controllers

A learned controller has to be shown competent at plain tracking before its obstacle behavior means anything, so this section places it beside tuned classical control on the no-obstacle task. The framing matters. On pure path tracking a classical controller is complete and verifiable, and for that task it is the sensible choice, so the aim here is not to beat it but to establish that the learned controller reaches comparable tracking competence. The classical controllers were tuned by differential evolution and validated on held-out seeds, and every controller faces the same pool as the learned agent. Tables 5.15 and 5.16 report the comparison.

Table 5.15: Forward lane-following controller benchmark on the evaluation pool. Tracking is saturated for every method, so completion cannot separate them here.

Controller	Completion (%)	Trailer CTE (m)	Max $\lvert\gamma\rvert$ (°)
TD3 (state, multiplicative)	$100.0_{\pm 0.0}$	0.463	8.2
Pure pursuit	100.0	0.151	5.2
PID	100.0	0.165	5.4
LQR	100.0	0.266	6.0
MPC	100.0	0.285	6.3

Table 5.16: Reverse lane-following controller benchmark on the evaluation pool. The learned policy nearly matches pure pursuit on completion and clears PID, while MPC, which receives an explicit path preview, is the informed upper bound.

Controller	Completion (%)	Trailer CTE (m)	Max $\lvert\gamma\rvert$ (°)
TD3 (state, multiplicative)	$95.9_{\pm 0.6}$	0.935	12.3
Pure pursuit	98.0	0.875	7.3
PID	87.3	0.852	15.7
LQR	95.9	0.619	10.1
MPC	97.7	0.845	8.1

Forward tracking is saturated, so completion cannot separate the methods and the comparison there is only about tracking quality. Reverse is more informative. The learned policy reaches completion comparable to pure pursuit and ahead of PID, which establishes the competence the obstacle results depend on, a policy that can actually stabilize and track the trailer on the unstable reverse task. This is a competence result rather than a claim of superiority. On the unfiltered pool pure pursuit completes 98.0% of the reverse layouts and the learned policy 95.9%, a small edge for pure pursuit on plain reverse tracking that is expected and is not the basis for the learned controller. The case for the learned controller does not rest on winning this tracking comparison. It rests on the integrated obstacle task of section 5.5, where pure pursuit alone is not a controller and the classical alternative is the full planner-and-tracker stack. In articulated control, success is not simply the lowest cross-track error, because hitch stability takes precedence over exact path following. A policy that holds the hitch angle inside a safe margin throughout a reverse maneuver is more successful than

one that shaves cross-track error but risks jackknifing, which is why the benchmark reports completion, trailer tracking, and articulation stability together rather than collapsing them into one score.

5.6.1 Information Parity between Learned and Classical Controllers

A controller comparison is only meaningful if the methods are given comparable information about the reference path. Table 5.17 summarizes what each method observes at a single control step on the no-obstacle task. The TD3 observation contains the instantaneous tractor and trailer tracking errors together with two path-curvature samples taken ahead of the trailer axle. This is a strict superset of the reference information used by pure pursuit and PID, which close the loop on the same instantaneous errors plus a single curvature sample at one lookahead point. The learned policy is therefore not at an information disadvantage relative to these baselines, if anything it is marginally favored, so the comparison against pure pursuit and PID is a fair, like-for-like test of control quality rather than one confounded by what each method is allowed to see.

Table 5.17: Reference-path information available to each method at a single control step on the no-obstacle task. The learned policy's observation is a strict superset of the information used by the pure-pursuit and PID baselines. Only MPC receives an explicit preview of the path ahead.

Method	Instantaneous errors (e_y, e_ψ)	Curvature lookahead	Full path preview
Pure pursuit	yes	1 pt ($\sim$10 m)	no
PID	yes	1 pt ($\sim$10 m)	no
TD3 (state/LiDAR/BEV)	yes	2 pts ($\sim$10, 20 m)	no
MPC	implicit	no	N=16/20 steps

MPC is the exception, and it is best read as an informed upper bound rather than a peer. It is supplied with an explicit preview of the path, a full reference trajectory over a receding horizon reconstructed from the same centerline the policy tracks, while the learned policy receives only the two scalar curvature samples. Where MPC attains lower cross-track error, part of that advantage is therefore attributable to horizon information the learned policy never sees, rather than to a superior control law alone. The pure-pursuit and PID comparisons are the information-matched test of whether a learned policy can equal classical

feedback control on the articulated tracking task, and closing the remaining gap to MPC by giving the policy an equivalent path-preview observation is identified as future work in chapter 7.

For obstacle scenarios the benchmark question changes, because the learned obstacle policy is solving a joint local-planning and control problem, including the stop/go decision, whereas the classical controllers are path trackers. The obstacle results are therefore not presented as directly comparable to controller-only baselines unless those baselines are given a matched planning layer and longitudinal policy.

5.7 Failure Modes and Deployment Relevance

Average metrics alone are insufficient for articulated-vehicle control, because the operational significance of a failure depends on how it occurs. The analysis therefore examines representative failure cases for each controller family, such as gradual corner-cutting, growing lateral oscillations, abrupt jackknife onset, and obstacle-clearance failures, rather than mean error alone. Trajectory overlays and hitch-angle traces are the natural diagnostics, because they reveal whether an error mode is geometric, dynamic, or perception-driven, and therefore which part of the pipeline a fix should target. The seed collapses behind the wide confidence intervals of section 5.3 are a case in point, since they are not a uniform degradation but a bimodal outcome where a policy either learns the task or falls into a degenerate behavior, which is visible in a trajectory trace in a way a mean completion rate hides.

This failure-oriented reading is also what connects the simulation study to deployment. A controller that attains a low mean cross-track error but exhibits rare catastrophic jackknife episodes is less deployable than one with slightly higher average error and consistent stability margins, which is why the reverse results are read through hitch stability rather than cross-track error alone. For the same reason controller latency and architectural modularity are weighed alongside tracking performance. Overall, the results position the learned controller as a credible peer to classical feedback control on the unstable reverse task, and the geometry-encoder recipe as the piece that makes high-dimensional perception trainable rather than merely available, which is what the obstacle task and eventual hardware deployment in chapter 6 require.

Chapter 6

Sim-to-Real Deployment

6.1 Overview

The preceding chapters developed and evaluated an articulated tractor-trailer controller entirely within a 2D simulation environment. This chapter describes the transfer of that controller to a physical robotic platform, constituting the sim-to-real component of this thesis. Unlike a conventional model-based stack, the deployed system does not re-implement a hand-derived tracking law on the vehicle. Instead, the simulation-trained reinforcement-learning (RL) policies are executed directly on hardware through a thin bridging layer that reproduces the observation and action interfaces the policies were trained against. The objective is to demonstrate that the vehicle model, the observation definitions, and the learned control logic established in simulation can be ported to real hardware with minimal algorithmic redesign, and to characterize the engineering challenges encountered during that transfer.

The deployment is built on a deliberately stripped-down Autoware stack. The standard Autoware localization and perception pipelines are retained essentially unchanged, while the behavior, planning, and control layers are replaced by a custom RL bridge that runs the forward and reverse policies. Two pieces of supporting infrastructure are described in detail because they are specific to the articulated, sensor-light platform: an onboard LiDAR-based hitch-angle estimator that recovers the articulation angle without any trailer-mounted sensor, and a Smith predictor [42] that compensates for actuator delay so that delay-free-trained policies behave correctly on a lagging physical actuator.

Figure 6.1: The physical articulated platform: an AgileX Hunter SE tractor coupled to the lab trailer through a revolute hitch at the tractor rear axle.

6.2 Hardware Platform

6.2.1 Tractor Vehicle

The physical tractor is an AgileX Hunter SE, a compact electric wheeled unmanned ground vehicle with an Ackermann steering geometry. The relevant mechanical parameters are a wheelbase of 0.65 m, a wheel radius of 0.15 m, and a maximum steering angle of 0.436 rad (25°). Vehicle actuation is commanded over a CAN bus on a 20 ms cycle. Speed commands are encoded as 16-bit signed integers in units of 1 mm/s (the commanded speed in m/s scaled by 1000) and steering commands as 16-bit signed integers scaled by approximately 1320 counts per radian (≈ 0.8 mrad per count).

6.2.2 Trailer

The trailer is coupled to the tractor through a revolute hitch joint located marginally behind the tractor rear axle, which was added to the tractor-trailer model used for training. The lab trailer has a wheelbase of approximately 2.8 m and a body width of roughly 0.33 m. The articulation angle γ, defined as the difference between the tractor and trailer headings

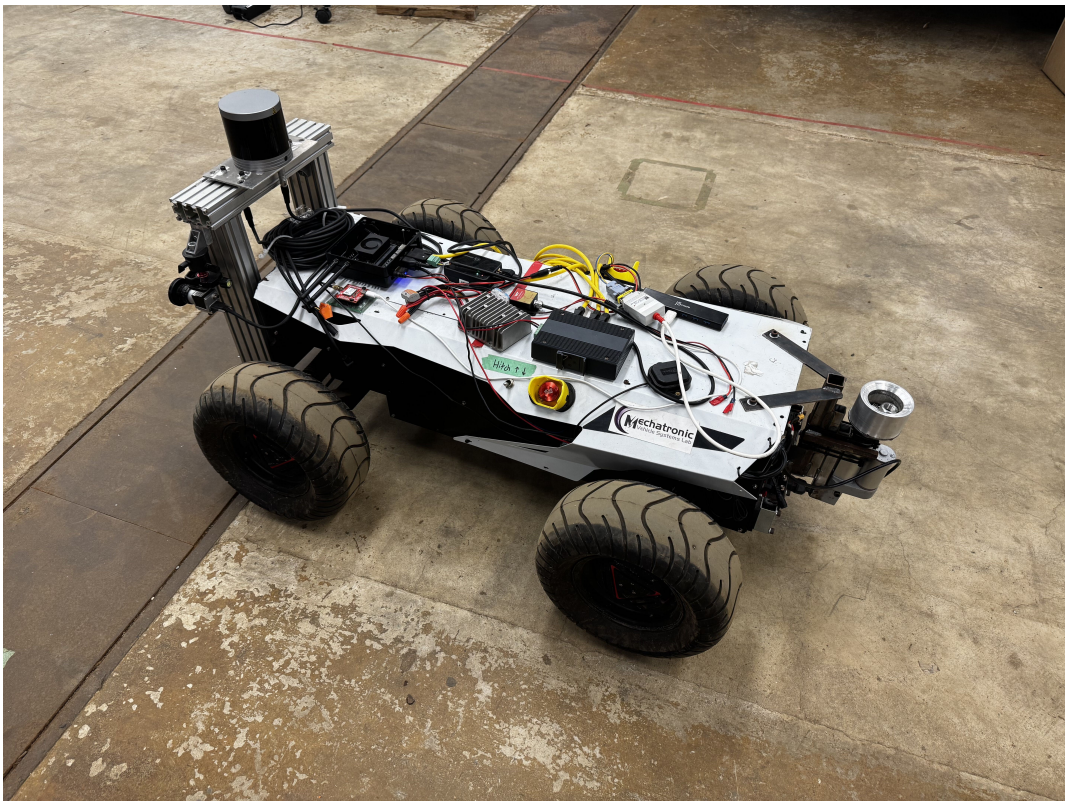


Figure 6.2: Onboard sensor suite: the front-mounted RoboSense Helios LiDAR, the Basler cameras, the u-blox GNSS antenna, and the WitMotion IMU.

($\gamma = \psi_{\text{truck}} - \psi_{\text{trailer}}$), is estimated onboard from the tractor's own LiDAR rather than measured by any trailer-mounted sensor. The procedure is detailed in Section 6.3.4. The estimate is published to the ROS2 network on the `/vehicle/trailer_state` topic. The control bridge consumes the most recent published value, with no staleness gating currently applied. Two distinct angular limits appear in the stack and should not be conflated. The onboard estimator clamps the published articulation estimate to $|\gamma| \leq 90°$, whereas the deployment trailer model defines a tighter operational limit of $|\gamma| \leq 80°$ (1.40 rad), itself below the 90° jackknife condition used to terminate simulation episodes.

6.2.3 Sensor Suite

The vehicle is instrumented with the following sensors:

- **LiDAR**: RoboSense Helios mechanical LiDAR mounted at the front of the tractor, publishing 3D point clouds on `/rslidar_points`. This single sensor is reused for three purposes: NDT-based localization [83], environment perception, and onboard hitch-angle estimation.

- **Cameras**: Six Basler Pylon industrial cameras (rear, rear-right, center, left, merging, and right) producing compressed color images. The cameras are not consumed by the LiDAR-based RL policies and are used for logging and monitoring only.
- **IMU**: WitMotion 6-DoF inertial measurement unit, with a gyroscope-bias estimator applied in the localization pipeline.
- **GNSS**: u-blox ZED-F9P receiver with RTK corrections, providing centimeter-level position estimates converted to MGRS coordinates for Autoware localization.
- **Wheel odometry**: Vehicle speed and steering-angle feedback recovered from CAN messages and fused with IMU data for dead-reckoning between GNSS and scan-matching updates.

6.3 Software Architecture

6.3.1 Middleware and Framework

The deployment stack runs on ROS2 Humble (Ubuntu 22.04) [84] and is integrated into the Autoware autonomous-driving framework [85]. Autoware provides modular localization, perception, planning, and control pipelines whose interfaces are defined by standardized ROS2 message types. As introduced in the overview, this deployment keeps the standard sensing, localization, and perception modules and replaces only the planning and control layers with the learned RL bridge.

6.3.2 Relationship to the Conventional Articulated Pipeline

For context, Autoware's conventional approach to articulated vehicles is fully model-based. A truck-trailer mode manager routes goals to a freespace planner (a hybrid A^* search [15] that respects the non-holonomic constraints of the articulated vehicle), and the resulting reference trajectory is tracked by a model-predictive or LQR lateral controller. This model-based pipeline exists within the wider project and represents the standard, well-understood baseline for trailer control. The relevant control background is reviewed in Chapter 2. The contribution of this thesis, however, is the learned controller, and the deployment described here therefore routes control through the RL bridge instead of the planner/MPC chain. The two are mutually exclusive at run time and are selected at launch. Figure 6.3 shows which modules are retained and which are replaced.

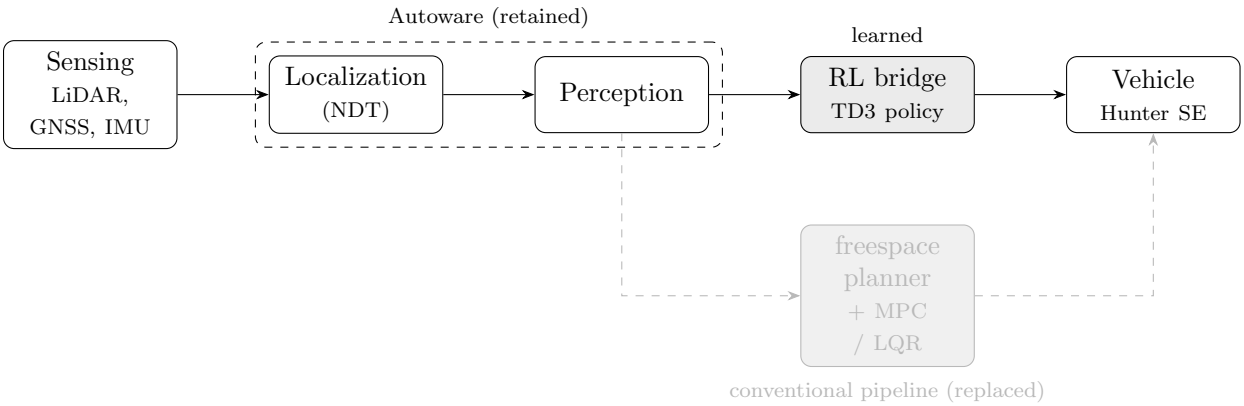


Figure 6.3: Deployment software stack. Autoware's localization and perception are retained unchanged, while the planning and control layers (the conventional freespace planner and MPC/LQR tracker) are replaced by the learned RL bridge. The two control paths are mutually exclusive and selected at launch.

6.3.3 RL Bridge Node Architecture

The learned control path comprises the following principal nodes:

`lane_reference_node` Loads the Lanelet2 map, selects the active lane from the ego odometry and the current goal, and publishes the lane centerline as a reference path together with a `drive_enabled` flag and a `drive_direction` flag (forward or reverse). This node supplies the path geometry against which the policy's tracking observation is computed.

`rl_bridge_node` The core of the deployment. It loads the trained forward and reverse TD3 policies (Stable-Baselines3 [72] / PyTorch [86] checkpoints) and runs a 10 Hz control loop. On each tick it assembles the policy observation in exactly the layout used during training, an 8-dimensional state vector (steering angle, articulation angle, tractor and trailer tracking errors, and look-ahead path curvature) concatenated with a 24-beam LiDAR range vector, queries the active policy deterministically, and converts its 2-dimensional action (a steering rate and a stop/go signal) into the steering-angle and speed commands expected by the vehicle interface, with the stop/go signal mapped onto the task's fixed speed or a halt exactly as the simulation interface layer does during training. Direction-dependent post-processing (action smoothing and kinematic rate scaling) is applied for the reverse policy, whose outputs are inherently more oscillatory than the forward policy's.

`hitch_angle_estimator` Estimates the articulation angle from the tractor LiDAR and publishes it on `/vehicle/trailer_state` (Section 6.3.4).

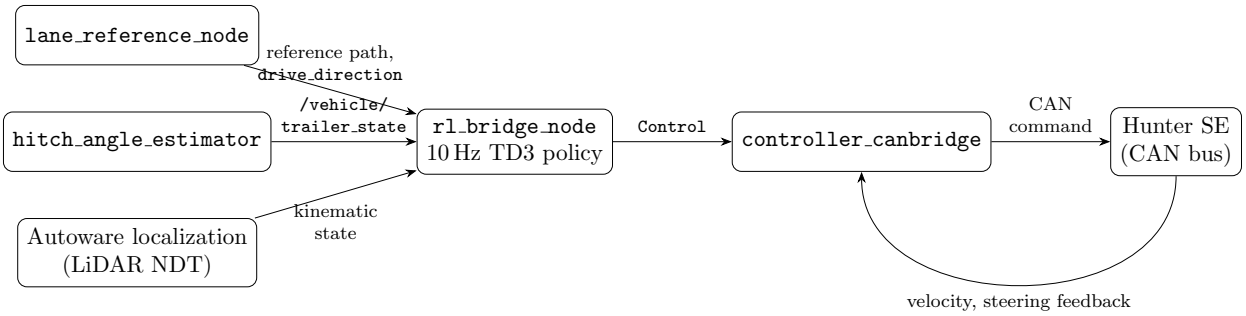


Figure 6.4: RL bridge dataflow. The `lane_reference_node`, the LiDAR-based `hitch_angle_estimator`, and Autoware localization supply the policy observation to `rl_bridge_node`, which queries the TD3 policy at 10 Hz and emits a control command that `controller_canbridge` translates into Hunter SE CAN frames. Vehicle velocity and steering feedback return through the same bridge.

`initialpose_shim` Bridges the RViz `/initialpose` message to the Autoware pose-initializer service, allowing the operator to seed localization interactively.

`controller_canbridge` Translates `autoware_control_msgs/Control` commands into Hunter SE CAN frames and parses vehicle feedback (velocity, steering angle) back into ROS2 topics.

The forward and reverse policies are loaded from separate checkpoints and selected according to the `drive_direction` flag, so that the same bridge handles both maneuvering regimes without retraining or reconfiguration.

Figure 6.4 shows how these nodes connect.

6.3.4 Hitch Angle Estimation

Because the trailer carries no articulation encoder and no trailer-mounted sensing, the articulation angle is recovered geometrically from the tractor's front LiDAR, which observes the front (and, at larger angles, the side) face of the trailer. The estimator operates on each incoming point cloud as follows.

Region of interest

Points are first cropped to a box in the LiDAR frame that brackets the expected location of the trailer face. The lateral extent of this region is shifted to track the face as the trailer swings: using the current filtered estimate γ, the lateral offset is set to

$$\Delta y = -\,D_{\text{face}}\,\sin\gamma, \tag{6.1}$$

Figure unavailable

LiDAR returns on the trailer face with the fitted rectangle used for articulation-angle estimation.

`figures/hardware/hitch_lidar_fit.png`

Figure 6.5: Onboard hitch-angle estimation from the tractor LiDAR: returns on the trailer face (points) with the fitted minimum-closeness rectangle, whose orientation gives the articulation angle γ.

where D_{face} is the approximate longitudinal distance from the hitch pivot to the face. This keeps the cropped points on the trailer face across the full range of articulation rather than only near $\gamma = 0$.

Rectangle fitting

Rather than fitting a single planar normal, the estimator fits the orientation of the trailer body to the cropped points using the minimum-closeness rectangle criterion of Zhang et al. [87]. For a candidate orientation θ, the points are rotated into axis-aligned coordinates (u, v) and scored by their closeness to the nearest edges of the bounding rectangle,

$$C(\theta) = \sum_i \min\big(d_i^u,\, d_i^v\big)^2, \qquad d_i^u = \min\big(|u_i - u_{\min}|,\, |u_i - u_{\max}|\big), \tag{6.2}$$

with d_i^v defined analogously. The fitted orientation minimizes $C(\theta)$ augmented by a dimension-prior penalty that favors rectangles consistent with the known trailer width and length:

$$\theta^\star = \arg\min_{\theta \in [0,\pi)} \; \big[\, C(\theta) + \lambda\, P_{\dim}(\theta) \,\big]. \tag{6.3}$$

This formulation is robust to whether only the front face, both the front and a side face (an "L" shape), or only a side face is visible, the dominant cause of failure for a naive single-face normal estimate at large articulation. The four 90° heading candidates implied by $\theta^\star$ are disambiguated by choosing the one closest to the previous filtered estimate and most consistent with the dimension prior, yielding a trailer longitudinal direction $\hat{\mathbf{d}}_t = (\cos\theta^\star, \sin\theta^\star)$.

Articulation angle and filtering

The raw articulation angle is the signed angle between the truck-forward axis $\hat{\mathbf{x}} = (1, 0)$ and $\hat{\mathbf{d}}_t$,

$$\tilde{\gamma} = \operatorname{atan2}\big(\hat{x}_1 \hat{d}_{t,2} - \hat{x}_2 \hat{d}_{t,1},\; \hat{\mathbf{x}} \cdot \hat{\mathbf{d}}_t\big), \tag{6.4}$$

to which a field-calibration sign s and offset γ_0 are applied and the result clamped to the operational range, $\gamma = \mathrm{clip}(s\,\tilde{\gamma} + \gamma_0,\, -\gamma_{\max},\, \gamma_{\max})$. The sign convention is chosen so that the published value matches Autoware's $\gamma = \psi_{\text{truck}} - \psi_{\text{trailer}}$. The raw per-scan estimate is then passed through a constant-angular-velocity Kalman filter [88] on the state $[\gamma,\ \dot{\gamma}]$, with Mahalanobis innovation gating to reject outlier scans, and a short moving-average is maintained in parallel as a diagnostic. The filtered angle and its rate populate the `hitch_angle` and `hitch_rate` fields of the published `TrailerState`.

This onboard, geometry-plus-filter estimator is the actual mechanism behind the `/vehicle/trailer_state` signal. It is not an external measurement. Its noise characteristics, and the way they are handled, are revisited in the gap analysis (Section 6.4.3).

6.3.5 Trailer State Estimation

Because the trailer carries no independent localization, its full pose is reconstructed from the tractor's localized kinematic state and the estimated articulation angle. Given the tractor's rear-axle position and heading $(\mathbf{p}_{\text{truck}}, \psi_{\text{truck}})$ from `/localization/kinematic_state`, the hitch-point position is

$$\mathbf{p}_{\text{hitch}} = \mathbf{p}_{\text{truck}} + d_h \begin{bmatrix} \cos\psi_{\text{truck}} \\ \sin\psi_{\text{truck}} \end{bmatrix}, \tag{6.5}$$

where d_h is the (small, near-zero) signed hitch offset relative to the tractor rear axle. The trailer heading follows directly from the articulation angle, $\psi_{\text{trailer}} = \psi_{\text{truck}} - \gamma$, and the trailer rear axle is located at

$$\mathbf{p}_{\text{trailer}} = \mathbf{p}_{\text{hitch}} - L_t \begin{bmatrix} \cos\psi_{\text{trailer}} \\ \sin\psi_{\text{trailer}} \end{bmatrix}, \tag{6.6}$$

where $L_t \approx 2.8\,\text{m}$ is the trailer wheelbase. This reconstruction mirrors the hitch-kinematics model derived in Chapter 3, confirming that the simulation formulation is directly reusable in the deployment stack, and it provides the trailer-referenced tracking errors that form part of the policy observation.

6.3.6 Actuator-Delay Compensation: Smith Predictor

The simulation trains the policies against an idealized actuator that responds instantaneously to commands, whereas the physical steering and drive systems exhibit a first-order lag of the order of 50–150 ms. A policy trained for instantaneous response will, when confronted with a delayed actuator, repeatedly over-correct and excite a lag-induced oscillation. To reconcile the two, a Smith-predictor-style compensator is implemented that lets the delay-free-trained policy act on a delay-compensated state estimate.

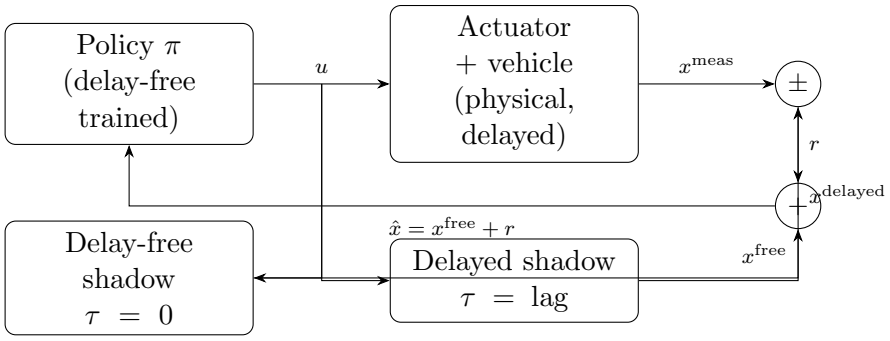


Figure 6.6: Smith-predictor compensator. The command u drives the physical vehicle and two kinematic shadows. The delayed shadow is subtracted from the measured state to form a residual r, which corrects the delay-free shadow to give the delay-compensated estimate $\hat{x} = x^{\text{free}} + r$ presented to the policy.

The compensator maintains two shadow instances of the same kinematic `StateSpace TractorTrailer` model used in simulation: a delay-free shadow with zero actuation time constant, and a delayed shadow whose time constant matches the measured hardware lag. On each control tick both shadows are advanced with the command currently propagating through the actuator pipeline. The discrepancy between the measured vehicle state and the delayed shadow forms a residual,

$$\mathbf{r}_k = \mathbf{x}_k^{\text{meas}} - \mathbf{x}_k^{\text{delayed}}, \tag{6.7}$$

which captures model mismatch beyond pure delay (integration scheme, ground-contact effects, friction). The state presented to the policy is the delay-free shadow corrected by this residual,

$$\hat{\mathbf{x}}_k = \mathbf{x}_k^{\text{free}} + \mathbf{r}_k, \tag{6.8}$$

so that the policy sees an effectively delay-free vehicle while the residual keeps the prediction anchored to reality. The shadows are re-synchronized to the measured state on reset and whenever a large state jump is detected. The compensator is implemented, enabled, and its shadow time constants are set from the AgileX's measured actuator lag. Combined with the explicit steering-rate limit it forms the primary mitigation for the actuation gap discussed in Section 6.4.2.

Figure 6.6 shows the compensator structure.

6.4 Sim-to-Real Gap Analysis

Transferring a simulation-trained policy to physical hardware introduces discrepancies between the simulated and real-world dynamics. The following gap sources were identified and addressed during the deployment process.

6.4.1 Vehicle Scale and Parameterization

The most fundamental difference between the simulation studies and the physical platform is one of scale. The simulation of chapter 3 is parameterized for a full-size shunt truck (table 3.1), whereas the deployment vehicle is the compact AgileX Hunter SE (wheelbase 0.65 m, maximum steering angle 0.436 rad), with under a quarter of the wheelbase and half the steering range of the reference terminal tractor. Rather than transferring the shunt-truck-scale policy onto the smaller vehicle and relying on it to generalize across scale, the deployment closes this gap by construction. The model structure of chapter 3 is retained unchanged and re-parameterized to the Hunter's wheelbase, steering limit, and speed range, and a fresh policy is trained against that re-parameterized model. Because the governing equations are identical and only the parameter set changes, the observation and action interfaces the policy was trained against are preserved, and the scale mismatch is removed before deployment rather than being left as a residual gap to be compensated at run time.

6.4.2 Actuation Dynamics

The simulation model assumes instantaneous steering response, whereas the physical vehicle exhibits a first-order steering lag introduced by the CAN command latency and the Hunter SE's internal steering servo. This is the highest-impact discrepancy for a policy trained on instantaneous actuation. It is mitigated by the Smith-predictor compensator of Section 6.3.6, which presents a delay-compensated state to the policy, and by a steering-rate limit applied to the commanded output, consistent with the $\dot{\delta}_{\max}$ constraint used during simulation training.

6.4.3 Hitch-Angle Sensing

In simulation, the articulation angle γ is available exactly from the vehicle state. On hardware it is estimated from LiDAR returns and is therefore subject to noise from sparse or partial returns on the trailer face, mis-segmentation of the region of interest, and momentary loss of the face at large articulation. These error modes are addressed within the estimator itself rather than by a downstream filter. The rectangle-fitting formulation (Equation 6.3)

is robust to partial face visibility, and the constant-angular-velocity Kalman filter with innovation gating rejects outlier scans and smooths the published signal. If the trailer face is lost entirely the estimator withholds new estimates, and the bridge continues from the last published value until the face is reacquired. No automatic disengagement on stale data is currently implemented.

6.4.4 Path Distribution

The policies are trained on smooth, spline-generated reference paths, whereas the deployment reference is the centerline of a Lanelet2 lane, which is piecewise-linear and may contain comparatively sharp vertices. Re-sampling the centerline does not fully reconcile the difference in path-curvature distribution between training and deployment, and this mismatch is a contributing factor to any residual tracking error observed on hardware. It is noted here as a characterization of the gap rather than something fully eliminated in the current deployment.

6.4.5 Jackknife Prevention

The simulation introduced a terminal episode condition when $|\gamma| > 90°$ but provided no explicit recovery mechanism, since the episode simply resets. Rather than adding a separate runtime supervisor on the vehicle, jackknife resilience is instilled during training, through two complementary measures. First, a recovery-scenario reset distribution spawns a fraction of reverse episodes already in the near-jackknife region (articulation magnitudes of roughly 0.35–0.80 rad, i.e. 20–45°), so the policy is repeatedly exposed to high-articulation states its nominal training distribution rarely visits and must learn to steer the trailer back toward alignment. Second, an anti-jackknife override is applied within the training environment. Whenever the articulation exceeds approximately 0.7 rad (40°), the policy's steering command is replaced by a maximum counter-steer that drives the articulation back toward zero. This prevents the policy from learning that oscillatory, near-jackknife behavior is survivable, since every dangerous excursion is corrected during training and chattering no longer pays off. No dedicated runtime recovery override is currently part of the deployed bridge. On hardware, jackknife avoidance therefore relies on the behavior learned through these measures together with the operational $|\gamma| \leq 80°$ limit.

6.4.6 Localization Dependency

The simulation provides ground-truth vehicle pose at every step. The hardware stack instead depends on the Autoware localization pipeline, which fuses RTK-GNSS, gyro-based odometry, and LiDAR NDT scan matching through an extended Kalman filter. GNSS outages or localization divergence therefore represent a failure mode that does not exist in simulation. The current deployment treats localization failure as an emergency-stop condition.

6.5 High-Fidelity Geographic Simulation

The 2D Pygame environment of chapter 3 is deliberately abstract. It captures the tractor-trailer kinematics and lane geometry but not the three-dimensional and photometric structure that the deployed perception and localization stack actually consumes. A complementary, higher-fidelity simulation environment was therefore built to exercise the full Autoware stack (LiDAR-based NDT localization, perception, and the RL bridge) inside a geographically faithful "digital twin" of a real operating site as a complement to the physical trials.

This environment is built on the Gazebo simulator [89], extending the open-source Rudis Laboratories georeferenced world-generation toolchain [90]. Real locations are reconstructed as textured 3D meshes authored in Blender [91], whose physically based materials are baked into texture maps for efficient real-time rendering. Building on that toolchain, the target distribution-center site was rebuilt in Blender directly from public geographic data, with both layers imported through the Blender GIS plugin. Building footprints were pulled from OpenStreetMap [92] and extruded to form the 3D building meshes, and these were draped over aerial orthoimagery from the ESRI World Imagery basemap, replacing the lower-resolution ground textures of the original toolchain with the higher-resolution ESRI imagery (fig. 6.7). Because the reconstruction is driven entirely by the OpenStreetMap footprints and the ESRI imagery layer, the same procedure generates a Gazebo world for any real distribution center from its coordinates alone, rather than being tied to a single hand-authored site. Each world is anchored to GIS coordinates through the SDF `spherical_coordinates` element, which fixes the simulation origin to a real latitude, longitude, and elevation. The templated world generator additionally computes the sun's azimuth and elevation from the site coordinates and a chosen UTC date and time, so that lighting and shadows match the real location and time of day. The `electrans` tractor and trailer models are then spawned into these worlds, so that the same articulated vehicle can be driven through a virtual replica of the target environment (figs. 6.8 and 6.9).

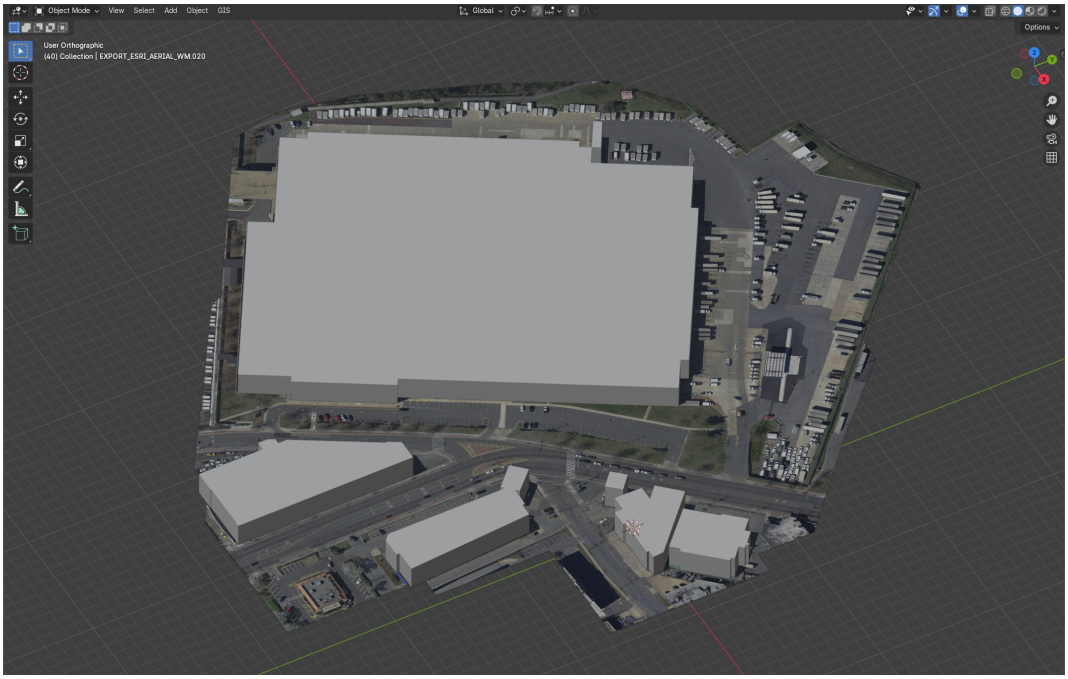


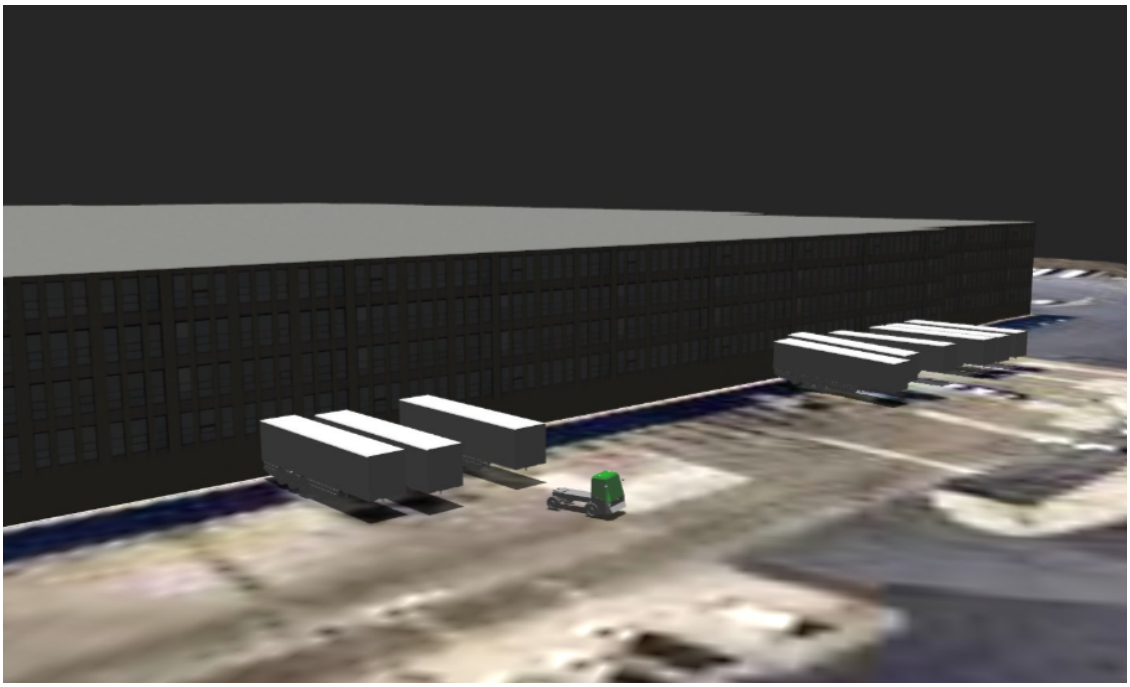
Figure 6.7: Reconstruction of the target distribution-center site in Blender. OpenStreetMap building footprints are extruded to form the 3D building meshes and draped over high-resolution aerial orthoimagery from the ESRI World Imagery basemap, both layers imported through the Blender GIS plugin. The same GIS-driven procedure reconstructs any real distribution center from its geographic coordinates.

Figure unavailable

Reconstructed distribution center exported to a georeferenced Gazebo world.

`figures/gazebo/gazebo_dc_world.png`

(a) The reconstructed site exported to a georeferenced Gazebo world.



(b) The `electrans` tractor-trailer spawned at a loading dock within the world.

Figure 6.8: The georeferenced site running in Gazebo. The exported world (a) preserves the real building geometry and aerial ground texture, and the `electrans` articulated vehicle (b) is spawned into it so that the same policies evaluated in simulation can be exercised against the full Autoware sensing and localization stack.

Because the world is georeferenced and rendered in three dimensions, this environment directly targets several of the sim-to-real gaps catalogued above. A LiDAR sensor model produces point clouds against real building and ground geometry, so NDT scan matching and the onboard hitch-angle estimator (section 6.3.4) can be exercised against realistic returns; the geographic anchoring lets the GNSS and map frames be reproduced consistently with the physical site; and the photorealistic rendering makes it possible to evaluate camera-based perception that the abstract 2D environment cannot represent. It is also the natural setting in which to prototype the infrastructure sensor nodes of section 6.6, since virtual LiDAR units can be placed at arbitrary surveyed locations in the georeferenced world. The environment and its GIS-driven world-generation pipeline are complete. What remains is

the quantitative characterization of policy behavior within it, which is identified as future work in chapter 7.

6.5.1 Articulated Vehicle Model

The articulated vehicle spawned into the georeferenced worlds is the `electrans` tractor-trailer, shown in fig. 6.9. It is defined in the conventional ROS/Gazebo manner as a URDF assembled from `xacro` macros, rather than as a hand-written SDF model. The tractor shares its entire `xacro` description with the AgileX Hunter model used elsewhere in the project. The same parameterized macros define an Ackermann steering geometry (a pair of revolute front-steering links bounded by the maximum steering angle), continuous-rotation rear wheels coupled by a mimic joint, the link inertials, and the Gazebo friction and `ros2_control` blocks. The two vehicles differ only in scale and in the body mesh. The Hunter's compact chassis mesh is replaced by a cab-over terminal-tractor body, and the model is scaled accordingly. Reusing a single `xacro` description in this way keeps the simulated tractor kinematically consistent with the physical Hunter platform on which the policies are deployed, while presenting the larger visual form of the terminal tractor this thesis targets.

The trailer is coupled to the tractor through a revolute hitch joint at the tractor rear axle, matching the hitch convention of the kinematic model in chapter 3. It is modeled as a single box body carried on one axle and reuses the same wheel meshes and inertial macros as the tractor, so that the complete articulated system is expressed through the shared macro library. Because the model is authored as standard xacro, it drops directly into both the Gazebo digital twin of this section and the wider Autoware simulation tooling without modification.

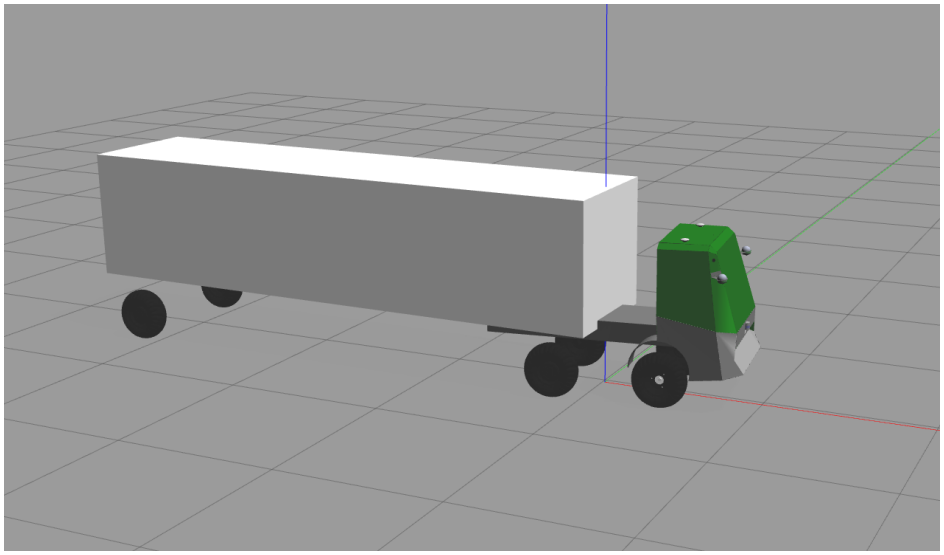


Figure 6.9: The `electrans` tractor-trailer model. The tractor reuses the AgileX Hunter `xacro` description (Ackermann front steering, continuous rear wheels, and `ros2_control` blocks), rescaled and given a cab-over terminal-tractor body mesh. The box trailer is attached through a revolute hitch joint at the tractor rear axle.

6.6 Infrastructure Sensor Nodes

A structural limitation of the onboard sensing arrangement is that the trailer occludes the region directly behind it. The tractor-mounted LiDAR cannot see past the trailer, and instrumenting the trailer itself is undesirable. It would add cabling, power, calibration, and a wireless link to an otherwise passive, interchangeable body, and would have to be repeated for every trailer. This is most acute during reverse maneuvering, precisely when visibility of the space behind the trailer matters most.

This limitation is addressed with off-vehicle infrastructure sensor nodes, fixed LiDAR units placed at surveyed vantage points around the operating area, each providing a bird's-eye view of obstacles (including the region behind the trailer) without any sensor being mounted on the trailer. Each node is calibrated into a shared global frame so that its detections can be expressed in the same coordinates as the onboard perception output and fused into Autoware's perception and occupancy-grid representation, yielding a persistent, occlusion-resilient view of the obstacles around the articulated vehicle that complements the limited onboard field of view.

This capability is implemented and running in the laboratory as the indoor-perception stack. The infrastructure sensing system described here was developed separately by Cao and colleagues in the laboratory [93], and this thesis discusses only its integration boundary

with the ROS2, Autoware, and RL-bridge deployment stack. Each infrastructure node runs Euclidean clustering on its own LiDAR returns to detect candidate objects. An offboard fusion pipeline transforms every node's detections into the shared global frame, merges detections that refer to the same physical object across nodes by their overlapping footprint polygons, and time-aligns the streams against each node's modeled latency before passing the merged detections to an extended-Kalman-filter multi-object tracker. The tracker emits a single stable, deduplicated obstacle list for the whole monitored area.

Because the laboratory sensor nodes and their perception stack run under ROS1 (Noetic) while the vehicle runs under ROS2 (Humble), the two are joined by a dedicated interface node. It subscribes to the ROS1 perception outputs and republishes the fused obstacle list over ROS2, where it is consumed by the vehicle's Autoware perception and occupancy-grid representation and thus reaches the RL bridge. Critically, this interface node runs offboard on the laboratory compute rather than on the vehicle. The vehicle's onboard Jetson therefore runs only ROS2, and never has to host a dual ROS1/ROS2 stack. The ROS1 dependency is confined entirely to the off-vehicle infrastructure, which both simplifies the embedded software image and keeps the version-sensitive ROS1 perception packages off the resource-constrained onboard computer.

In laboratory operation, the infrastructure nodes recover the region directly behind the trailer that the onboard LiDAR cannot observe. Quantitative coverage and latency characterization are left for the final deployment evaluation.

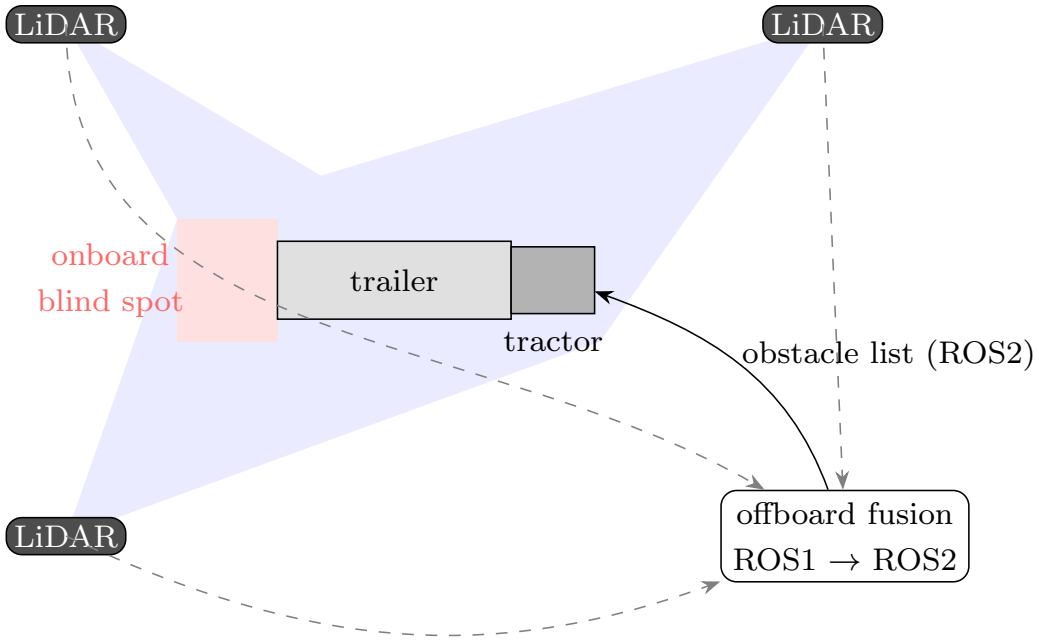


Figure 6.10: Infrastructure sensor-node architecture: fixed off-vehicle LiDAR units give a bird's-eye view of obstacles around the articulated vehicle, covering the region behind the trailer that the onboard sensors cannot observe. Their detections are fused and tracked offboard and delivered to the vehicle over ROS2, so the onboard computer runs only ROS2.

6.7 Deployment Results

6.7.1 Trial Protocol and Metric Capture

The deployment was exercised on the laboratory site as a point-to-point lane-following maneuver run in both directions. Each trial asked the vehicle to travel from a fixed start pose A to a fixed goal pose B along a surveyed lane centerline, a segment of roughly 23 m taken from a 60.5 m lane published in the map frame by the lane-reference node. Forward and reverse trials cover the same physical segment, the difference being that the tractor leads in the forward trials and the trailer leads in the reverse ones. The relevant policy ran through the bridge at 10 Hz with the Smith predictor of section 6.3.6 enabled and the articulation estimate supplied by the onboard LiDAR estimator of section 6.3.4. The bridge commanded a constant speed in each direction, 0.59 m/s forward and 0.25 m/s in reverse, so the policy controlled steering rate alone and the trials isolate lateral tracking. A safety operator supervised every trial and could stop the vehicle at any point.

Nine trials were recorded on 2026-08-05. One was a stationary bench session in which no velocity was ever commanded, and four were reverse trials ended by the safety operator

Figure unavailable

Photo of the articulated platform during a deployment trial at the laboratory site.

`figures/hardware/deployment_trial.jpg`

Figure 6.11: The articulated platform during one of the deployment trials at the laboratory site.

before reaching point B. This section reports the four trials that completed the maneuver under policy control, two in each direction. The reverse failures are not incidental and are taken up in section 6.7.3, where they turn out to share a single mechanism with the behavior seen in the trials that did succeed.

Each trial was recorded to a rosbag carrying the localization estimate (`/localization/kinematic_stat`, the articulation estimate (`/vehicle/trailer_state`), the steering feedback (`/vehicle/status/steering_`, the issued control command (`/control/command/control_cmd`), the reference centerline and its drive-enable and direction flags, and the bridge's own observation vector and raw policy action. Statistics are reported over the policy-controlled segment of each trial, defined as the ticks where the lane-reference node reports drive-enabled in the intended direction and the bridge is commanding a non-zero velocity in that direction. The segment closes at the first zero-velocity command sustained for one second, so that any operator stop and subsequent manual recovery drive is excluded from the numbers.

The reported errors are geometric rather than read back from the policy input, because the observation the policy sees has passed through the Smith predictor and a frame mirroring step and is therefore not a measurement of where the vehicle actually was. The tractor reference point is the localization pose, which is also the hitch location, and the trailer axle is reconstructed from the articulation estimate as $p_t = p - L_t\,[\cos(\psi - \gamma), \sin(\psi - \gamma)]^\top$ with the deployment trailer length $L_t = 2.8\,\mathrm{m}$. Cross-track error is the signed perpendicular offset from that point to the reference polyline, and heading error is the body heading relative to the path tangent taken along the direction of travel. As a consistency check, this reconstruction was correlated against the bridge's own logged observation over every trial, and the two agree with a magnitude correlation of at least 0.98 on articulation, tractor error, and trailer error, confirming that the offline reconstruction and the online observation describe the same motion.

6.7.2 Tracking Performance

Table 6.1: Lane following on the physical platform: the four successful A-to-B trials recorded on 2026-08-05, two in each direction, over the same surveyed lane segment. Statistics cover the policy-controlled segment of each trial and are computed geometrically from the localization estimate against the reference centerline, independently of the policy's own observation. The final column is the fraction of control ticks at which the commanded steering rate sat on its saturation limit.

Trial	Driven (m)	Speed (m/s)	Tractor CTE (m)		Trailer CTE (m)		$\lvert\gamma\rvert$ (deg)		$\dot{\delta}$ sat (%)
			mean	max	mean	max	mean	max	
Forward									
F1 (10:07:56)	23.6	0.59	0.045	0.10	0.223	1.05	9.1	38.7	0
F2 (10:18:02)	23.6	0.59	0.078	0.16	0.263	1.03	9.4	39.0	0
Reverse									
R1 (12:09:51)	22.6	0.25	0.964	2.39	0.687	1.91	20.9	51.6	80
R2 (12:45:49)	22.6	0.25	1.102	2.35	0.764	1.86	21.8	51.1	79

Table 6.1 summarizes the four trials and fig. 6.12 shows the tractor tracks against the reference lane. All four reached point B under policy control, and no trial jackknifed or came close to it. Beyond that the two directions behave so differently that they are best read separately.

Forward tracking transferred essentially intact. The tractor held a mean cross-track error of 0.045 and 0.078 m across the two trials, with worst-case deviations of 0.10 and 0.16 m, which is tracking at the scale of the localization noise rather than of the controller. The trailer followed at 0.22 and 0.26 m mean error and never left the lane corridor in either trial. Mean heading error was 1.3° for the tractor and 3.8 to 4.0° for the trailer. The two forward trials agree with each other closely enough that their tracks overlay in fig. 6.12, so the behavior is repeatable and not a single lucky run. Almost all of the forward error is concentrated in the final curve, where the trailer error rises from a mean of 0.10 m over the first 18 m to 0.60 m over the last 5 m and articulation reaches 39°, which is the expected cost of cutting a corner rather than a tracking failure.

Reverse is where the transfer degrades. The trailer held a mean cross-track error of 0.69 and 0.76 m with worst excursions near 1.9 m, roughly three times the forward figure, and spent 11% of the maneuver outside the 1.41 m lane corridor. The tractor error is larger still, at 0.96 and 1.10 m mean against 2.39 and 2.35 m peak, and is the one metric where the

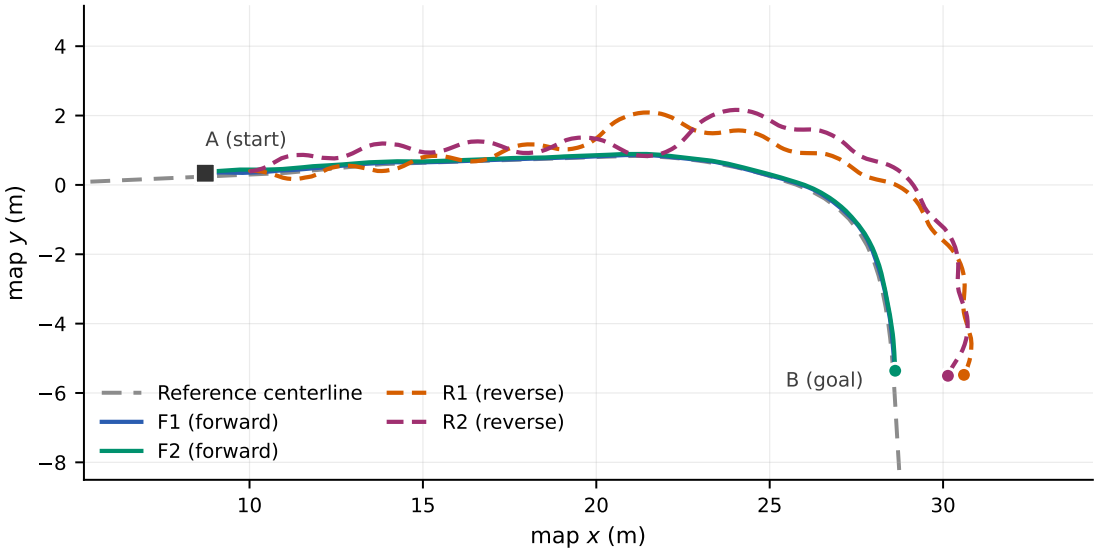


Figure 6.12: Tractor tracks for the four successful deployment trials against the surveyed lane centerline. Solid tracks are forward trials, dashed tracks are reverse. The two forward tracks are nearly coincident and largely hide one another. The forward trials sit on the reference lane throughout, while the reverse trials bow outward by up to two meters before converging near point B.

tractor is worse than the trailer. That inversion is the expected signature of a reverse policy rewarded on trailer placement, which will carry the tractor wide to put the trailer where the path wants it. Mean articulation is 21° against 9° forward, and peak articulation reaches 51.6°, still comfortably inside the 80° operational limit and the 90° jackknife condition.

6.7.3 Articulation Limit Cycle

The difference between the two directions is not simply that reverse tracks less accurately. It is that the reverse trials exhibit a behavior the forward trials do not show at all, and that simulation does not reproduce: a sustained oscillation in the articulation angle. Figure 6.13 makes the contrast plain. In the forward trials the articulation trace is smooth and drifts slowly, moving off zero only for the final curve. In the reverse trials the hitch angle swings back and forth continuously, with a median peak-to-peak amplitude of 44 to 46° and a spatial wavelength of 2.4 to 2.6 m, roughly one trailer length per cycle, sustained from the start of the maneuver to the end.

The third panel identifies the mechanism and rules out the obvious alternative explanations. In the forward trials the commanded steering rate never once reaches its 0.436 rad/s saturation limit. In the reverse trials it sits on that limit for 79 to 80% of the control ticks, so the reverse policy is not regulating articulation smoothly but bang-banging the steering

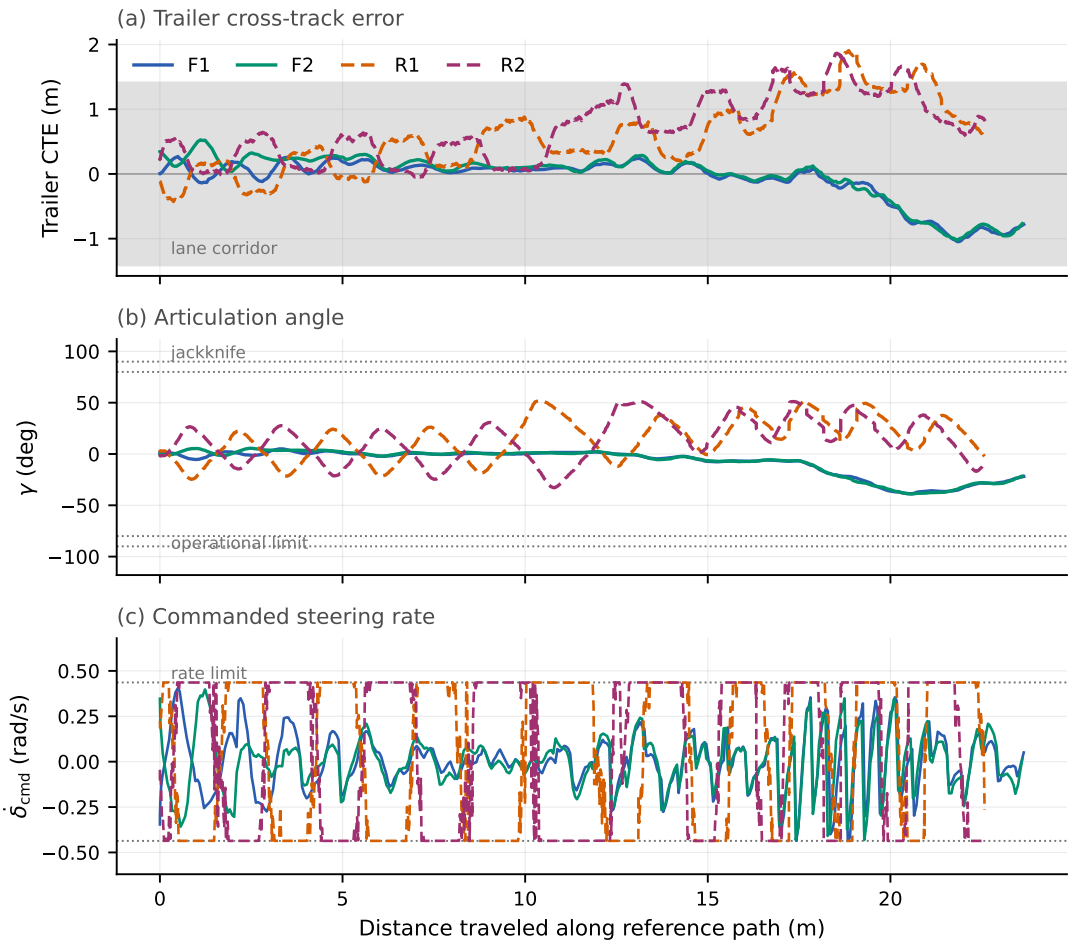


Figure 6.13: The four successful deployment trials against distance traveled along the reference path. (a) Trailer cross-track error, with the lane corridor shaded. (b) Articulation angle against the operational and jackknife limits. (c) Commanded steering rate against its saturation limit. Solid traces are forward trials, dashed traces are reverse. The reverse trials oscillate continuously and hold the steering rate on its limit, while the forward trials do neither.

between its rate limits. Since both directions run on the same vehicle, the same actuator, the same articulation estimator, and the same bridge on the same day, neither the hardware nor the sensing can by itself account for the difference. What differs is the plant. Forward articulation dynamics are self-stabilizing, so estimate noise and actuation lag are damped out, while reverse articulation dynamics are unstable and the same disturbances are amplified into the limit cycle.

This is the behavior expected of a delay-free-trained policy driving a lagging actuator, and it is the residue of the gap identified in section 6.4.2. The Smith predictor removes enough of the loop delay that the vehicle remains stable and no trial jackknifed, but it

does not remove all of it, and on the unstable reverse plant what is left closes a bang-bang limit cycle rather than settling. Articulation-estimate noise from the LiDAR rectangle fit of section 6.4.3 contributes on the same terms, since the policy acts on a noisy γ at every tick. That both effects are invisible in the forward trials is consistent with this reading rather than an argument against it, because forward articulation dynamics damp exactly the disturbances that reverse dynamics amplify.

On a straight lane the limit cycle is largely benign, since the oscillation is symmetric and averages out, and the reverse trials hold the trailer inside the lane corridor through the straight section. It stops being benign at the curve, where the policy must hold a sustained articulation offset rather than swing around zero while having already spent most of its steering-rate authority on the oscillation. This is also where the four reverse trials excluded in section 6.7.1 were ended. Each tracked the straight section normally for 11 to 13 m, then lost the trailer outward as the lane began to curve, reaching trailer errors between 3.9 and 4.2 m before the operator intervened. The reference curvature at that point saturates the 0.3 m^{-1} clip applied to the curvature observation, so the policy also loses the ability to distinguish the sharpest part of the turn from a merely tight one exactly where it most needs to anticipate. Reverse tracking on this route is therefore best described as succeeding on the straight and marginal at the curve, with the margin decided by where in its oscillation cycle the vehicle happens to enter the turn.

6.7.4 Comparison with Simulation

The simulation benchmarks of tables 5.15 and 5.16 give the learned policy a 0.463 m mean trailer cross-track error forward and 0.935 m in reverse, with peak articulation of 8.2° and 12.3°. Those distances are not directly comparable to the hardware figures, because the simulated vehicle is at Tesla Model S scale with a 10 m trailer while the laboratory platform carries a 2.8 m trailer. Normalized by trailer length, the simulated policy tracks to 0.046 trailer lengths forward and the deployed policy to 0.087, and in reverse to 0.094 against 0.26. Forward transfer therefore costs roughly a factor of two in relative tracking error, and reverse transfer roughly a factor of three.

The articulation comparison is the more informative one, because it is a pure angle and needs no scale correction. Forward peaks at 8.2° in simulation and 39° on hardware, but that excursion is confined to the final curve and the trace is otherwise smooth. Reverse peaks at 12.3° in simulation and 51.6° on hardware, and unlike the forward case the hardware trace oscillates for the entire maneuver. Simulation reproduces nothing resembling the limit cycle of fig. 6.13 in either direction.

Taken together the two directions localize the sim-to-real gap more sharply than either could alone. The forward trials transfer at near-simulation quality on the same vehicle, the same actuator, the same estimator, and the same bridge that the reverse trials use, which rules out the vehicle model, the observation definitions, the action interface, and the bridge implementation as the source of the reverse degradation. What is left is the interaction between the residual actuation lag and articulation-estimate noise, both anticipated by sections 6.4.2 and 6.4.3, and the open-loop instability of reverse articulation dynamics that turns those small imperfections into a sustained oscillation. The deployment data now quantifies what that interaction costs.

What these trials establish should be read narrowly. They show that policies trained entirely in the 2D simulator drive a physical articulated vehicle along a real surveyed lane, at a scale and with a trailer geometry never seen in training and without any retraining on hardware, and that forward lane following transfers at a quality close to simulation. They do not establish reverse tracking at a quality suitable for unsupervised operation, and with two trials per direction they cannot characterize a completion rate. Closing the remaining gap is a matter of modeling the actuator lag and the articulation-estimate noise during training rather than compensating for them at deployment, which is taken up in chapter 7.

6.8 Summary

This chapter demonstrated that the tractor-trailer control architecture developed in simulation can be ported to a physical platform by executing the trained RL policies directly through a bridging layer, rather than by re-deriving an analytical controller on the vehicle. The deployment retains Autoware's standard localization and perception augmented with the infrastructure sensor node obstacle list while replacing planning and control with an RL bridge that runs the forward and reverse policies at 10 Hz over the same observation and action interfaces used in training. The principal engineering contributions are this RL bridge and its integration into a stripped-down Autoware stack, the onboard LiDAR hitch-angle estimator that recovers the articulation angle by rectangle fitting and Kalman filtering without any trailer-mounted sensor, and the Smith-predictor compensator that reconciles delay-free-trained policies with a lagging physical actuator. The gap analysis confirms that the simulation captures the dominant dynamics relevant to controller design, while identifying actuation delay, hitch-angle sensing, path-distribution mismatch, and localization reliability as the primary points of divergence between the simulated and physical systems. The deployment trials of section 6.7 bear this out and sharpen it. Forward lane following transferred at close to simulation quality, holding the tractor within 0.16 m of the reference

lane without any retraining, which shows that the vehicle model, the observation definitions, and the action interface all carried across correctly. Reverse ran on that same validated stack and degraded, sustaining a bang-bang articulation limit cycle that simulation does not exhibit and that leaves the maneuver marginal wherever the lane curves. The two gaps the analysis flagged first, actuation delay and articulation sensing, are therefore also the two that now limit deployed performance, and they bite only where the plant is open-loop unstable. Finally, the infrastructure sensor nodes, implemented and running in the laboratory, provide a route to occlusion-resilient perception behind the trailer without adding sensors to the trailer itself.

Chapter 7

Conclusion

7.1 Summary of Findings

[Section content omitted for review.]

7.2 Limitations

[Section content omitted for review.]

7.3 Future Work

[Section content omitted for review.]

Appendix A

Classical Controller Implementation

The deep reinforcement-learning policy is benchmarked against four classical path-tracking controllers. This appendix documents their implementation: the exact control laws, the forward and reverse variants, the way each controller was tuned, and the harness used to evaluate them. The four controllers are chosen to span the classical control spectrum, from a purely geometric law through to constrained optimal control with preview:

- **Pure Pursuit**, a geometric proportional law;
- **PID**, classical error feedback with an integral term;
- **LQR**, optimal static linear feedback on the error dynamics;
- **Kinematic LTV MPC**, optimal constrained control with preview.

All four are implemented in the batched simulator (`tractor_trailer_rl_cupy`) so that they run inside the same vectorized environment as the learned policies. Crucially, every controller emits the same action as the RL agent, a bounded steering rate, so the comparison between classical and learned control is made on identical terms.

A.1 Common framework

A.1.1 Observation layout and shared action mapping

Each controller is a function of the observation vector already produced by the environment. For the trailer tasks this vector is

$$o = [\, s,\ \gamma,\ e_y,\ e_\psi,\ e_{y,t},\ e_{\psi,t},\ \kappa,\ \ldots\,], \tag{A.1}$$

where s is the current steering angle, γ the hitch (articulation) angle, and the error pairs (e_y, e_ψ) and $(e_{y,t}, e_{\psi,t})$ are the cross-track and heading errors of the tractor and of the trailer respectively; κ is the reference path curvature at the lookahead point. The trailing entries carry additional task-specific channels (for example LiDAR returns) that the classical controllers do not use.

Every controller computes a desired steering angle δ_{des} and then converts it to the environment's steering-rate action through a common saturating map,

$$\dot{\delta} = \text{clip}\left(\frac{\delta_{\text{des}} - s}{\Delta t},\ -\dot{\delta}_{\max},\ +\dot{\delta}_{\max}\right), \tag{A.2}$$

where Δt is the simulation step and $\dot{\delta}_{\max}$ the steering-rate limit. Because the learned policy also outputs $\dot{\delta}$, and eq. (A.2) applies to all four baselines identically, the classical and learned controllers act through exactly the same interface.

A.1.2 Forward and reverse tracking

Each controller has a forward and a reverse variant, reflecting a structural asymmetry of the articulated vehicle. Driving forward, the tractor leads and the trailer follows passively, so it suffices to regulate the tractor errors (e_y, e_ψ) towards zero. Driving in reverse, the trailer leads and the coupling inverts. The hitch angle γ becomes open-loop unstable (small articulation errors grow and the vehicle jackknifes), so the reverse controllers regulate the trailer errors $(e_{y,t}, e_{\psi,t})$ and must additionally include an explicit hitch-stabilizing action. This asymmetry is why the reverse variants below are not simply the forward laws with a sign change, and why their gains had to be tuned separately (appendix A.6).

A.1.3 Vehicle parameters

The benchmark uses the full-size shunt-truck configuration. The parameters that enter the controllers are summarized in table A.1. Here L_1 is the tractor wheelbase, L_2 the effective trailer length (hitch to trailer axle), and V the fixed longitudinal speed held constant throughout the fixed-speed path-tracking experiments. The large length ratio $L_2/L_1 \approx 4.2$ is what makes the reverse hitch dynamics difficult to stabilize.

Table A.1: Shunt-truck parameters used by the classical controllers.

Symbol	Quantity	Value
L_1	Tractor wheelbase	2.95 m
L_2	Trailer length (hitch to axle)	12.5 m
V	Fixed longitudinal speed	5.0 m/s
Δt	Simulation step	0.1 s
$\delta_{\max}$	Steering-angle limit	0.87 rad ($\approx 50°$)
$\dot{\delta}_{\max}$	Steering-rate limit	25°/s

In the reverse variants the speed enters with a sign, $V < 0$. Because the environment's tractor yaw response to steering does not itself flip sign in reverse, the along-path kinematics use the magnitude $|V|$ while the yaw dynamics retain the signed V. This distinction is carried consistently through the LQR and MPC models below.

A.2 Pure Pursuit

The pure-pursuit controller [18] is a fixed proportional geometric law. Forward, it combines a curvature feed-forward with proportional cross-track and heading feedback on the tractor errors,

$$\delta_{\text{des}} = -\big(k_{\text{ff}} \arctan(L_1 \kappa) + k_y\, e_y + k_\theta\, e_\psi\big). \tag{A.3}$$

Reverse, it regulates the trailer errors and adds an explicit hitch-stabilizing term $k_{\text{hitch}}\, \gamma$ to counter the open-loop-unstable articulation,

$$\delta_{\text{des}} = k_{\text{hitch}}\, \gamma + k_{\text{ff}}\, \kappa + k_y\, e_{y,t} + k_\theta\, e_{\psi,t}. \tag{A.4}$$

Both laws are stateless functions of the observation, so the controller is fully vectorized across the parallel environments. Besides serving as a benchmark, the tuned pure-pursuit controller doubles as the reference ("teacher") signal for the guided reward and as a feasibility oracle. A path that a well-tuned pure-pursuit controller can complete is deemed geometrically feasible.

A.3 PID

The PID controller extends the proportional law with an integral term that removes steady-state cross-track offset. The integral is accumulated per environment and clamped to prevent

windup, and it is reset at episode boundaries. Forward,

$$\delta_{\text{des}} = -k_{\text{ff}}\,\kappa - K_p\,e_y - K_i\int e_y - K_d\,e_\psi + K_{p,t}\,e_{y,t}, \tag{A.5}$$

and reverse (regulating the trailer errors, with the hitch stabilizer),

$$\delta_{\text{des}} = k_{\text{hitch}}\,\gamma + k_{\text{ff}}\,\kappa + K_p\,e_{y,t} + K_i\int e_{y,t} + K_d\,e_{\psi,t}. \tag{A.6}$$

The derivative gain K_d multiplies the heading error, so it acts as a heading-damping term rather than a numerical time derivative. The integrator uses the anti-windup update

$$I \leftarrow \text{clip}\big(I + e\,\Delta t,\ -I_{\max},\ +I_{\max}\big), \tag{A.7}$$

where e is the regulated cross-track error (e_y forward, $e_{y,t}$ reverse). Because the controller carries this per-environment state, the evaluation loop passes a termination mask so that I is zeroed for any environment that resets on the current step.

A.4 LQR

The LQR baseline is the canonical optimal-linear controller, static feedback $u = -Kx$ on a linear model of the tracking-error dynamics, plus a curvature feed-forward [67]. The error state is

$$x = [\,e_{\text{lat}},\ e_{\text{head}},\ \gamma\,]^\top, \tag{A.8}$$

the lateral and heading error of the leading unit (tractor forward, trailer reverse) together with the hitch angle. The input is the steering angle $u = \delta$. The continuous error dynamics $\dot{x} = A_c x + B_c u$ are direction-specific. Forward,

$$A_c = \begin{bmatrix} 0 & V & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\dfrac{V}{L_2} \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ \dfrac{V}{L_1} \\ \dfrac{V}{L_1} \end{bmatrix}, \tag{A.9}$$

and reverse, where the leading trailer progresses along the path at speed $|V|$ while the yaw dynamics keep the signed V,

$$A_c = \begin{bmatrix} 0 & |V| & 0 \\ 0 & 0 & \dfrac{V}{L_2} \\ 0 & 0 & -\dfrac{V}{L_2} \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ 0 \\ \dfrac{|V|}{L_1} \end{bmatrix}. \tag{A.10}$$

The pair (A_c, B_c) is discretized by zero-order hold at the step Δt to give (A_d, B_d). With state and input weights $Q = \mathrm{diag}(q_{\mathrm{lat}}, q_{\mathrm{head}}, q_\gamma)$ and R, the infinite-horizon gain follows from the discrete algebraic Riccati equation. Its stabilizing solution P yields

$$K = \left(R + B_d^\top P B_d\right)^{-1} B_d^\top P A_d. \tag{A.11}$$

Because the speed is constant, K is computed once at construction and then applied to every environment, so the controller stays fully vectorized. The command adds a steady-state feed-forward that holds the vehicle on a constant-curvature arc: a feed-forward steer $\delta_{\mathrm{ff}} = L_1 \kappa$ and a hitch reference $\gamma_{\mathrm{ref}} = \mathrm{sign}(V)\, L_2\, \kappa$, giving

$$\delta = \delta_{\mathrm{ff}} - K\,(x - x_{\mathrm{ref}}), \qquad x_{\mathrm{ref}} = [\,0,\ 0,\ \gamma_{\mathrm{ref}}\,]^\top, \tag{A.12}$$

which is then saturated to $\pm\delta_{\max}$ and mapped to a steering rate through eq. (A.2). The reverse weights place a heavy penalty on control effort and hitch deviation, which is what damps the otherwise unstable reverse plant.

A.5 Kinematic LTV MPC

The final baseline is a linear time-varying model-predictive controller [6] that adds a finite preview horizon and explicit input constraints. The MPC uses the same error state $x = [\,e_{\mathrm{lat}}, e_{\mathrm{head}}, \gamma\,]^\top$ as the LQR but plans over an N-step horizon.

A.5.1 Prediction models

Forward, the tractor leads, so the MPC tracks the tractor errors with the steering angle δ as the input, giving the same (A_c, B_c) as the forward LQR model (A.9). Reverse, servoing the trailer pose with the unstable hitch dynamics inside the optimization causes the long shunt trailer to oscillate and jackknife. The reverse controller therefore uses a flatness / virtual-

input reformulation (following the Autoware trailer LTV-MPC): the QP plans the trailer as a stable bicycle driven by a virtual trailer steer δ_T, and the hitch enters only through an inner loop that has been closed to the stable first-order dynamics $\dot{\gamma} = K(\delta_T - \gamma)$,

$$A_c = \begin{bmatrix} 0 & |V| & 0 \\ 0 & 0 & \dfrac{V}{L_2} \\ 0 & 0 & -K \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ 0 \\ K \end{bmatrix}. \tag{A.13}$$

The reference curvature enters both models as a measured disturbance through $E_c = [\,0,\ -|V|,\ 0\,]^\top$. After the QP returns the virtual input, an analytic inner map realizes it on the physical tractor and supplies the anti-jackknife counter-steer,

$$\delta_f = \arctan\!\left(\frac{L_1}{L_2}\,\operatorname{sign}(V)\,\sin\gamma + \frac{L_1 K}{|V|}\,(\delta_T - \gamma)\right). \tag{A.14}$$

The $1/|V|$ factor is what flips the feedback sign in reverse and produces the stabilizing counter-steer.

A.5.2 Condensed quadratic program

The continuous models are discretized by zero-order hold and the state trajectory is condensed onto the input sequence, so that the optimization variable is the vector of N future steering inputs. With the stacked weight $\bar{Q} = I_N \otimes Q$, a control penalty R, an input-rate penalty R_d, and the first-difference operator $T = I - I^{(-1)}$ (so that T forms Δu), the cost Hessian is

$$H = 2\big(B_u^\top \bar{Q}\, B_u + R\, I + R_d\, T^\top T\big), \tag{A.15}$$

where B_u is the condensed input-to-state map. The problem is subject to a steering-rate box on the increments, $|\Delta u| \le \dot{\delta}_{\max}\,\Delta t$, and a steering box $|u| \le \delta_{\max}$. It is solved per environment with OSQP. The rate-penalty term $R_d\,T^\top T$ is weighted heavily in reverse so the controller does not chase cross-track measurement ripple. The overall per-step procedure is summarized in algorithm 1.

Algorithm 1 Kinematic LTV MPC step (per environment)

Precompute (once): discretize (A_c, B_c, E_c) by ZOH; build the condensed maps A_x, B_u, E_w and the constant Hessian H from (A.15).
On episode reset: clear the warm-start input $u_{\text{prev}} \leftarrow 0$.
Read $x_0 = [\, e_{\text{lat}}, e_{\text{head}}, \gamma \,]$ and curvature κ from the observation.
Form the linear cost term from x_0, the previewed curvature disturbance, and u_{prev}.
Solve the QP with OSQP subject to the steering-rate and steering boxes.
if solver fails **then** $u \leftarrow u_{\text{prev}}$ ▷ safe fallback
else $u \leftarrow$ first element of the optimal input sequence
end if
$u_{\text{prev}} \leftarrow u$
$\delta \leftarrow u$ (forward) **or** apply inner map (A.14) to obtain δ_f (reverse).
return steering rate from (A.2).

Only the first input of the optimal sequence is applied (receding horizon). The solved input is retained as the warm start for the next step and, like the PID integrator, is reset at episode boundaries. If the solver fails to converge, the controller falls back to the previous command.

A.6 Gain-tuning methodology

The controllers above are defined by their gains and weights: the proportional and feed-forward gains of pure pursuit, the PID gains, the LQR state/input weights, and the MPC horizon and cost weights. Rather than report a specific set of numbers, which are particular to one vehicle configuration, the point of interest is the procedure used to obtain them, which is the same for every controller.

Each controller is tuned by a bounded random search (a gain sweep) run directly against the simulator [94]. On each iteration a candidate gain set is sampled uniformly from hand-set ranges, the controller is rolled out over a batch of parallel environments, and the candidate is scored by the lexicographic key

$$\big(\text{completion rate},\ -\,\text{mean cross-track error}\big), \tag{A.16}$$

so that reliably completing the path is maximized first and, among gains that complete, the one with the tightest tracking is preferred. The best candidate is retained. This search is run separately for each controller and each direction, because the forward and reverse plants

differ so strongly.

The reverse variants in particular could not simply inherit the gains tuned on the small laboratory vehicle. On the shunt truck those gains destabilize the vehicle, and several gains (notably the hitch-stabilizing terms) change sign. This is a direct consequence of the open-loop-unstable reverse hitch dynamics and the large trailer-to-tractor length ratio $L_2/L_1 \approx 4.2$ discussed in appendix A.1.2. The reverse controllers were therefore re-tuned from scratch for the target geometry.

A.7 Evaluation harness

All four controllers are evaluated by a single harness that reuses the same batched environment as the reinforcement-learning policies, so that the classical baselines and the learned policy are measured on identical paths under identical dynamics. The harness runs both driving directions and, for each controller, records three per-episode metrics:

- **Completion**, the fraction of episodes that reach the goal (the mean success flag);
- **Trailer cross-track error**, the mean absolute lateral error of the trailer over the episode, a measure of tracking accuracy;
- **Peak hitch angle**, the largest articulation angle reached, a measure of how close the vehicle came to jackknifing (the jackknife threshold is 1.4 rad).

The harness handles the stateful controllers by passing a per-environment termination mask into the controller at each step, so the PID integrator (A.7) and the MPC warm start are reset exactly when an environment restarts. Evaluating every controller in the same loop, on the same unfiltered evaluation pool, is what makes the reinforcement-learning comparison in the results chapter a fair, like-for-like one. The measured numbers are reported there rather than in this appendix.

Appendix B

Full Hyperparameter Tables

Table B.1: TD3 Hyperparameters

[**Omitted for review**]
Table content omitted for review.

Appendix C

Additional Plots

[Section content omitted for review.]

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